

ESSENTIAL MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Similarity between objects – Enlargement – Scale factors (linear, area and volume); negative scale factor will not be included – Application of similarity and enlargement
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify properties of similar figures, - determine the centre of enlargement and the linear scale factor of similar figures, - draw the image of an object given the centre of enlargement and the linear scale factor, - determine the area scale factor of similar figures, - determine the volume scale factor of similar objects, - relate linear scale factor to area and volume scale factor of similar objects, - apply similarity and enlargement to real-life situations, - appreciate the use of similarity and enlargement in real-life situations.
Core Competencies	– Digital literacy: The learner develops the skills of connecting and interacting with digital technology when using digital resources to learn more on the use and application of similarity and enlargement. – Creativity and Imagination: the learner acquires networking skills when drawing images of given objects under a particular enlargement scale factor.
Core Values	– Responsibility: The learner takes care of objects collected from the immediate environment.
Science & Engineering Practices	– Developing and using models: learners draw enlargement images and build similar cubes, cuboids and cylinders (e.g. different sizes of Kericho tea tins) to represent scale relationships. – Analyzing and interpreting data: learners measure corresponding sides, areas and volumes of similar objects and tabulate the ratios to find patterns. – Using mathematics and computational thinking: learners compute linear, area and volume scale factors and use k, k^2 and k^3 to predict outcomes. – Constructing explanations: learners justify why doubling the length of an object does not double its area or volume.

	– Engaging in argument from evidence: learners defend their predictions about cost and quantity using measurement evidence. – Asking questions and defining problems: learners add questions to the Driving Question Board as they investigate.
Pertinent & Contemporary Issues (PCIs)	– Environment and technology: the learner acquires the skill of sorting in managing the environment after collecting and sorting the objects and containers.
Career Connections	– Photographer/Photojournalist: enlarging and reducing photos while keeping proportions the same. – Graphic designer and sign writer: scaling posters, billboards along Mombasa Road and logos. – Architect and building technician: reading scale drawings and building plans. – Cartographer and surveyor: using map scales in Survey of Kenya maps. – Hospitality and catering: scaling recipes and choosing container sizes (e.g. sufuria or mandazi trays) for bigger events. – Media technologist: resizing images and video on screens.
Focus for Lessons	Learners investigate how shapes and objects can be made larger or smaller while keeping their shape, and discover how lengths, areas and volumes change by k , k^2 and k^3 . The unit moves from sorting similar everyday objects, to drawing enlargements, to relating scale factors, and ends with a project making similar containers from local materials. Each lesson follows predict, observe, explain, Driving Question Board and model building.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object? KICD KEY INQUIRY QUESTIONS: 1. How is similarity and enlargement applied in real life?
Anchoring Phenomenon	A mandazi vendor in Kisumu sells small, medium and large sufurias of githeri that look exactly the same shape. The large sufuria is twice as tall and twice as wide as the small one and is priced at only twice the small price. Yet customers who buy the large sufuria say it feeds far more people than twice as many, and the vendor's cooking oil used to paint the outside costs more than double. Students observe the three sufurias, measure them, and find that doubling the length does not simply double the amount of food or paint. This is puzzling because everything looks doubled, yet capacity and surface do not behave as expected.
Storyline Thread	Lesson 1: Students meet the three sufurias phenomenon, predict which will feed more people, and collect and sort objects and containers from their surroundings into similar and non-similar sets, noting properties (equal corresponding angles, proportional sides). Initial DQB and model started. Lesson 2: Properties of similar figures: students measure corresponding sides and angles of pairs of objects and cut-outs, compute ratios, and decide which are similar. Model updated with the idea of constant ratio. Lesson 3: Enlargement and centre of enlargement: students trace an object and its image on plain paper, join corresponding points to locate the centre, and find the linear scale factor from measured sides. Lesson 4: Drawing images: given a centre of enlargement and a linear scale factor (positive only), students draw images of shapes on plain and squared paper and check them against measurements; digital zoom and shrink tools are used to compare. Lesson 5: Area scale factor: students compute the ratio of image area to object area for enlarged shapes and discover that it is the square of the linear scale factor. Model revised to explain paint and surface. Lesson 6: Volume scale factor: students compare the volumes of cubes and cuboids of different sizes (e.g. flour tins), discover k^3, and explain why the large sufuria holds much more githeri. Lesson 7: Relating linear, area and volume scale factors: students solve problems moving between k, k^2 and k^3 with real contexts (map scales, posters, containers) and revise the model to predict costs and capacities. Lesson 8: Project and application: students make similar containers (cubes, cuboids, cylinders) from local materials, document the

	process with photos or video, return to the sufuria phenomenon, and finalise the model and DQB.
Supporting Phenomena	– A passport photo enlarged to a poster for a Nairobi billboard keeps the same face shape, but the area of paper needed grows far faster than the width. – A map of Lake Victoria on a 1:50,000 scale shows a small shape, but the real lake covers a huge area, so a 1 cm length is not 1 cm² of area. – Two cube-shaped maize flour tins, one with sides twice the other, hold much more than twice the flour. – A shadow of a tree in the Rift Valley and a metre stick form similar triangles, letting learners find the tree's height without climbing it.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute lessons)): Meet Mama Achieng's Three Sufurias: Which Feeds More, and What Does 'Similar' Mean?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon of Mama Achieng's three sufurias at Kibuye Market, Kisumu, so that learners make first predictions about size and servings, and to introduce the idea that similar objects have the same shape through sorting real containers into similar and non-similar sets.
Knowledge	By the end of the lesson, the learner should be able to identify properties of similar figures (equal corresponding angles and corresponding sides in the same ratio) by sorting objects and containers collected from the immediate environment into similar and non-similar sets.
Skills	Predicting and justifying with reasoning (NGSS SEP: Asking Questions and Defining Problems); measuring lengths with a ruler and string; calculating simple ratios; sorting and classifying objects by shape properties; communicating ideas with peers; developing and using an initial model (NGSS SEP: Developing and Using Models).
Attitudes	Responsibility: the learner takes care of objects collected from the immediate environment. Curiosity about why doubling lengths does not simply double the amount. Respect: the learner listens to others' opinions while sorting and discussing in groups. Appreciation of the use of similarity and enlargement in real life. PCI (Environment and technology): the learner acquires the skill of sorting in managing the environment after collecting and sorting the objects and containers.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	This is the launch lesson. Learners meet the puzzle that the large sufuria is 2 times as tall and 2 times as wide as the small one and costs only twice as much (KSh 120 vs KSh 60), yet customers say it feeds far more than twice as many people. They make first predictions, learn what 'similar' means by sorting objects, and start the Driving Question Board (DQB) and an initial model. Later lessons will test similarity by measurement (Lesson 2), introduce enlargement and scale factor, and finally explain capacity and paint area. Final capacity and paint figures are NOT revealed in this lesson.
Safety Notes	Learners walk carefully when collecting objects and do not bring broken glass, sharp-edged or rusty tins; the teacher checks all objects before use. Flatten or smooth any cut tin edges with tape. Handle real sufurias with care (no hot food or hot water in class; use empty sufurias or cardboard models). Keep walkways clear of bags and objects. Wash hands after handling collected containers. Return or recycle all objects responsibly at the end.

B. LESSON OVERVIEW

The teacher tells the story of Mama Achieng, the mandazi and githeri vendor at Kibuye Market in Kisumu, who sells githeri in three sufuria sizes. Learners are shown three real or cardboard-model sufurias (S: height 12 cm, diameter 16 cm, KSh 60; M: height 18 cm, diameter 24 cm, KSh 90; L: height 24 cm, diameter 32 cm, KSh 120). They notice the large sufuria is 2 times as tall and 2 times as wide as the small one and costs only twice as much, yet customers say it feeds far more than twice as many people, and the oil paint on the outside costs more than double. Learners first predict individually, then in groups of 4 they measure the sufurias and compare, and collect and sort containers and objects (tins, cups, plates, books, photo frames, mandazi cutters, bottle tops, sheets of paper) into similar and non-similar sets, noting equal corresponding angles and proportional sides. Whole-class discussion builds a working definition of similar. The class opens a Driving Question Board with the driving question and their own questions, and starts a first model (a drawing and sentence) to explain why capacity and paint might not just double. No final capacity or paint figures are given: the puzzle stays open for later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners gather for the story of Mama Achieng at Kibuye Market, Kisumu. They examine the three sufurias S, M and L (real or cardboard models) and the dimension card: heights 12, 18, 24 cm; diameters 16, 24, 32 cm; prices KSh 60, 90, 120. Working alone first, each learner writes in their notebook: (1) Which sufuria will feed more people and roughly how many servings of githeri will each hold if one serving is a full plate? (2) Is the large sufuria 'just a bigger copy' of the small one? Why do you think so? (3) If the large sufuria is twice as tall and twice as wide, will it hold exactly twice as much githeri? They draw the three sufurias side by side and record a number prediction (for example, 'S feeds __, M feeds __, L feeds __ people'). Then they share with a partner and groups of 4 record the range of predictions on a chart. They also predict whether the oil to paint the outside will double.	Hold up the small and large sufuria and say: "Mama Achieng at Kibuye Market sells githeri in these three sizes. The large one is twice as tall and twice as wide as the small one, and she charges twice the price: KSh 60 and KSh 120. Customers say the large one feeds far more than twice as many people. Does that make sense to you?" Do NOT give any capacity or paint figures. Display the dimension card. Say: "Write your own prediction first. No talking for 2 minutes." Give individual WAIT TIME of 10-15 seconds after asking: "If you were the customer, which would you buy and why?" before anyone speaks. Circulate and note 3 different predictions (for example, 'twice as much', 'three times', 'much more'). Cold-call 4 learners from different groups to share their number prediction WITHOUT telling them if they are right; say "Thank you, I am recording that. We will test it." Record all predictions on the board in a tally. Ask: "What reasoning are you using to decide?" and call on 2 more learners.	Predict-then-Justify (Prediction Gallery): Learners commit to a prediction with a reason before any instruction. Public tally of the class predictions makes disagreement visible, creating a need to know. This surfaces the common misconception that doubling lengths doubles everything, which later lessons will test against evidence.	Observe whether learners (a) write a specific numerical prediction for servings rather than leaving it blank, (b) give a reason, even an informal one such as 'it is bigger in every direction', and (c) draw the three sufurias in a roughly proportional way. Tally how many predict 'exactly double' versus 'more than double'. Listen for the idea 'same shape, bigger size' as a prior knowledge indicator. Record on a checklist the 6-8 learners who give no reason to follow up later.
Observe Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor	Part A (about 15 min): In groups of 4, learners measure sufuria S, M and L with a ruler and string (height and diameter, using string around the base to compare), and record in a table: height, diameter, price. They calculate and record the ratios M:S and L:S for height and for diameter (12 : 18 : 24 and 16 : 24 : 32, so 1 : 1.5 : 2 each time) and compare the price ratios (60 : 90 : 120). They notice every	Say: "Measure carefully and record in your table. Ratio means comparing two lengths by dividing." Remind learners of ratio from Junior School: "What does 12 : 24 simplify to?" Cold-call 3 learners. Do NOT calculate capacity or paint area. If a group starts asking about volume, say: "Good question. Put it on a sticky note for our board. Today we only look at lengths and shapes." For	Sorting and Classifying with Evidence (Claim-Evidence): Learners sort first by looking, then test the claim with measurement of angles and side ratios. Contrasting cases (two rectangles with equal angles but different side ratios) show that equal angles alone is not enough and side ratios must also match. This builds the two-part idea of similarity from the learners' own evidence.	Look for groups that (a) measure lengths accurately (within about 2 mm) and record in a table, (b) correctly compute ratios for sufuria S, M, L and notice height and diameter ratios match, (c) place the trap objects (tall thin vs short wide rectangle) in the correct set using side ratios and not only appearance, (d) use words such as 'angles are equal' and 'sides are in the same ratio'. Use an

factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	length of the large sufuria is 2 times the small one, and that the base and the rim are circles in all three, and the sides are straight and upright in all three (angles at the rim and base look the same). Part B (about 25 min): Learners collect objects and containers from the classroom and school compound (before the lesson if possible) such as tins, cups, plates, books, exercise books, photo frames, football cards, a set of different-sized mandazi cutters, bottle tops, sheets of A4 and A3 paper, ID cards, and rectangular flour packets. In their groups they sort them into two sets: 'look the same shape, just different size' and 'do not look the same shape'. They then check their sets by (a) comparing corresponding angles using the corner of a sheet of paper (all right angles in rectangles, same circle shape) and (b) measuring two lengths on each object and checking whether the ratios match (for example, length : width of a small and a large exercise book). They record in a Similar / Not similar table with a column 'What I noticed about angles and sides'.	sorting, say: "Decide which objects are enlargements or reductions of each other in shape. Test your decision with measurement." Circulate and ask each group: "What is the same about the shapes in this set? What changed?" and "Show me the angles on these two objects." Deliberately place two traps among the objects: a tall thin rectangle and a short wide rectangle (both rectangles but not similar), and a square and a rectangle. Ask: "Both have four right angles. Are they the same shape? How can you tell?" Give WAIT TIME of 10-15 seconds before taking answers. Ask groups to take care of the objects and to put them back neatly. Remind: "Responsibility: handle each object carefully and check for sharp edges."		observation schedule: tick if the group tests with both angles and sides. Common errors to note: judging by 'looks similar' only; thinking 'all rectangles are similar'; comparing differences (add 2 cm) instead of ratios (multiply by 2).
Explain Phase  VIDEO: Using similarity to estimate ratio between side lengths <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners return to their desks with their sorting tables. In a whole-class discussion, each group presents one similar set and one non-similar set and gives their evidence. Together the class builds a working definition on a poster: 'Similar objects have the same shape: corresponding angles are equal and corresponding sides are in the same ratio.' Learners then apply this to the sufurias: they explain in pairs why S, M and L are similar to	Lead a structured talk: "Group 3, tell us one set you made and how you know the objects are similar." Ask the rest: "Do you agree or disagree? What evidence did you see?" Use the talk moves: "Who can add on?" and "Who can say it in different words?" Cold-call 5 learners (one per group, different from the presenters). Highlight the trap cases: "A tall thin rectangle and a short wide rectangle both have four right angles. What is different?" WAIT TIME 10-15	Claim-Evidence-Reasoning Talk with Revisiting Predictions: Learners use the sorting evidence to build a shared definition of similar, then apply it to the sufurias. Returning to their earlier predictions with new knowledge shows what they now know (the sizes are similar with all lengths 2 times) and what remains unexplained (capacity and paint), which motivates the rest of the unit.	Check whether learners can state both properties of similar figures in their own words and give an example from the sorting. Look in notebooks for a correct sentence such as 'S and L are similar because the height and diameter of L are both 2 times those of S and the shapes are the same.' Use a quick thumbs check: 'Are two rectangles always similar?' Expect most to say no with a reason. Note any learner who writes 'similar means equal' or who cannot

 EXERCISE: Scale drawings <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	each other (circular base and rim, same upright sides, all lengths in the ratio 1 : 1.5 : 2) and write a short explanation in their notebooks. They also revisit their earlier predictions and discuss: 'Now that we know the three sufurias are similar and the large one is 2 times the small one in every length, is it still reasonable to say it holds exactly twice as much?' They record what they are unsure about, for example: 'Length doubles, but what happens to the space inside?' and 'What happens to the area of paint?' They do NOT calculate capacity or paint area today.	seconds before each response. Write the learners' own wording on the poster, then tidy it to the KICD wording: equal corresponding angles and sides in proportion. Say: "Corresponding means matching: the height of S matches the height of L." Connect to the phenomenon: "Mama Achieng's three sufurias are similar. Every length of the large one is 2 times the small. Customers still say it feeds far more than twice as many. What puzzles you?" Do NOT confirm or reveal the capacity or paint figures. Say: "We will find out by testing, not by guessing. Keep your question for the board." Correct misconception that similar means identical or congruent: "Similar means the same shape, not the same size."		explain why a tall thin rectangle is not similar to a short wide one, and plan a short follow-up in Lesson 2.
Driving Question Board (DQB) Creation  VIDEO: Scale drawings <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Similarity & transformations <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The class opens the Driving Question Board on a wall chart or large manila paper titled 'Why is the large sufuria worth more than twice the githeri?' The teacher posts the driving question: 'Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object?' Each learner writes at least one question on a sticky note or card about the sufurias or about similar objects, for example 'How much more does L hold?', 'Why does paint cost more than double?', 'Is there a rule to predict this?', 'Are all enlargements like this?', 'How can I draw a bigger copy of a shape?' Groups cluster their notes by theme (similar shapes, area, capacity, price and paint cost, drawing enlargements) and stick them on the board. The	Say: "A Driving Question Board holds our questions. We will come back to it every lesson and move questions to 'answered' when we have evidence." Read the driving question aloud. Say: "Write what you wonder. One question per note. Use your own words." WAIT TIME 10-15 seconds, then give 3 minutes of silent writing. Cold-call 4 learners to read a question and ask: "Which part of Mama Achieng's problem does your question come from?" Group similar questions and ask: "Which of these questions can we answer by measuring? Which need us to find a rule?" Add questions of your own if learners miss them (for example, 'Is the price fair for the customer? For Mama Achieng?'). Do NOT answer questions today. Say: "Great question. Where do you think we will look for the answer?" Photograph the board to	Driving Question Board (question clustering): Learners turn confusion into public questions and organise them into themes. This gives purpose to the later lessons, lets learners see how their questions will be answered, and gives the class ownership of the investigation.	Collect and read the sticky notes. Look for questions that go beyond 'how big' to 'why' and 'what is the rule', which show conceptual curiosity. Count how many learners wrote a question and how many link to area, capacity or price. Use a checklist to note learners who wrote only factual or off-topic questions and prompt them with: 'What do you wonder about the space inside, or the paint outside?'

	class agrees on 2-3 'big questions' to return to in each lesson. They also add the three sufuria dimension card and the tally of predictions beside the board.	keep a record. Appoint two learners as board monitors (responsibility).		
Model Building Phase  VIDEO: Corresponding parts of congruent triangles are congruent <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Equations with Ratios and Proportions (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	In groups of 4, learners create Version 1 of the class model. On a sheet of manila paper they (1) draw the small and large sufurias side by side with the measurements labelled (height 12 cm and 24 cm; diameter 16 cm and 32 cm), (2) mark which things double (height, diameter, price) with arrows, (3) write 'We think capacity will be ___ times because ___' and 'We think paint area will be ___ times because ___', using their own words and drawings, and (4) write one sentence beginning 'The three sufurias are similar because ...' Groups display their models in a gallery walk and add one sticky-note comment on another group's model beginning 'I wonder...' or 'I agree because...'. Learners then keep a personal Summary Table in their notebooks with three columns: What we observed, What we learned, What we can explain so far. Models stay on the wall and will be revised in later lessons.	Say: "A model is our best explanation so far. It can be wrong. We will revise it as we get evidence." Provide the template with blanks. Circulate and ask: "What are you assuming about how capacity changes? Where did that come from?" Do not correct number predictions. Instead say: "Mark it as a prediction. We will test it." Cold-call 3 groups to explain their model in 30 seconds each. WAIT TIME 10-15 seconds after asking: "What would you need to find out to know if your model is right?" Run a 5-minute gallery walk with the instruction: "Leave one 'I wonder' note on at least two models." Close with an exit ticket: "In one sentence, what does 'similar' mean? Which part of Mama Achieng's puzzle do you most want solved?" Collect exit tickets. Preview: "Next lesson we will test similarity more carefully with rectangles and with the sufuria measurements." Remind learners to return all objects to the collection box.	Initial Model Building and Gallery Walk: Learners externalise their current thinking in a labelled diagram plus words, which makes their ideas visible and testable. Gallery comments introduce peer critique, and the model becomes the base that will be revised in every later lesson as evidence on k, k squared and k cubed is gathered.	Check each group's model for (a) correct labelled measurements, (b) a clear statement about what doubles, (c) a prediction with a reason (even if the prediction is 'exactly double'), and (d) a correct use of the word similar. Use the exit ticket: success is a sentence that mentions same shape with equal corresponding angles and sides in the same ratio. Sort tickets into three piles (complete, partial, misconception such as 'similar means equal') to plan the start of Lesson 2.

D. TEACHER REFLECTION

After the lesson, consider: Which predictions did learners make about servings and paint, and how many thought the large sufuria holds exactly twice as much? Did learners test similarity with both angles and side ratios, or only by appearance? Which trap objects (tall thin vs short wide rectangle) caught the most learners? Did any learners confuse similar with congruent or with 'equal'? Were all groups able to take measurements accurately, and did all learners take part in sorting and questioning? Which learners wrote strong DQB questions and which need prompting? Did the Kenyan context (Mama Achieng, Kibuye Market, githeri) increase interest and participation? What materials or time need adjusting for Lesson 2, where side ratios will be tested more carefully with rectangles A, B and C?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that the small, medium and large sufurias have the same shape: the height and diameter of the large one are both 2 times those of the small one (heights 12, 18, 24 cm; diameters 16, 24, 32 cm), while the price is KSh 60, 90 and 120. We also observed that some objects look the same shape in different sizes and others do not, even when both have right angles.
What did I learn?	We learned that similar objects have the same shape: their corresponding angles are equal and their corresponding sides are in the same ratio. Equal angles alone are not enough, because a tall thin rectangle and a short wide rectangle are not similar.
How does this explain the phenomenon?	So far we can explain that the three sufurias are similar because every length of the large one is 2 times the small one. We cannot yet explain why the large sufuria seems to feed far more than twice as many people, or why the paint costs more than double. Those questions are on our Driving Question Board and in our first model.

LESSON 2 (80 minutes (2 x 40-minute lessons)): Properties of Similar Figures: Is the Ratio the Same Everywhere?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners test whether pairs of figures are similar by measuring corresponding sides and angles and calculating side ratios, discovering that similar figures have equal corresponding angles and a constant ratio of corresponding sides. They then show that the sufuria sizes S, M and L share one constant ratio for height and diameter.
Knowledge	By the end of the lesson the learner should be able to identify properties of similar figures: corresponding angles are equal and corresponding sides are in the same (constant) ratio. Learners know that one unequal ratio is enough to show two figures are not similar. They know that the ratio of image length to object length is the linear scale factor k (formally developed in Lesson 3).
Skills	Measuring lengths to the nearest 0.1 cm and checking angles with a protractor or set square; calculating and comparing ratios; sorting figures as similar or not similar using evidence; Science and Engineering Practices: Analysing and Interpreting Data, Using Mathematics and Computational Thinking, Constructing Explanations, Engaging in Argument from Evidence.
Attitudes	Responsibility: the learner takes care of the cut-outs and measuring tools collected from the immediate environment. Respect and unity: the learner listens to the opinions of peers while checking measurements together. Appreciation of the use of similarity in real life, such as Mama Achieng's three sufuria sizes. Core competencies: Critical thinking and problem solving (interpretation and inference when comparing ratios); Communication and collaboration (group measuring and sharing of evidence). PCI: Life skills (decision making) when choosing between a fair and an unfair comparison.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	Lesson 1 sorted containers into similar and non-similar sets and made first predictions about the sufurias. Lesson 2 gives the sorting a test: measure corresponding sides and angles, then calculate ratios. Learners find that the ratio of corresponding sides stays constant for similar figures, and that sufuria S, M and L have height and diameter ratios of $1 : 1.5 : 2$. This is the idea of constant ratio, and it is added to the class model. It prepares Lesson 3, where this constant ratio is used as the scale factor of an enlargement. The capacity and paint puzzle is not yet solved: this lesson shows that every length is multiplied by the same factor, but not yet what happens to area and volume.
Safety Notes	Use blunt-tipped scissors and rulers; walk, do not run, while moving around the room with cut-outs. Keep measuring tools and cardboard sufuria models off the floor to prevent tripping. If real aluminium sufurias are used, check for sharp rims and handle them with two hands. Pick up all paper offcuts at the end of the lesson.

B. LESSON OVERVIEW

Mama Achieng at Kibuye Market in Kisumu sells githeri in three sufuria sizes that all look the same shape. The large one is twice as tall and twice as wide as the small one and costs KSh 120 against KSh 60, yet customers say it feeds far more than twice as many people. In Lesson 1 learners sorted objects into similar and non-similar sets. Today they ask how to check, rather than guess, that two shapes are really similar. Learners first predict which of three cut-out rectangles (A 4 cm by 6 cm, B 6 cm by 9 cm, C 6 cm by 8 cm) are similar to A. They then measure all sides and angles, and calculate the ratios $6/4 = 1.5$ and $9/6 = 1.5$ for B, and $6/4 = 1.5$ but $8/6 = 1.33$ for C. They discover that B is similar to A but C is not, even though one ratio matches. They then test the sufurias: heights 12, 18, 24 cm and diameters 16, 24, 32 cm give ratios $1 : 1.5 : 2$ for both. Learners explain why the sufurias are similar, record questions on the DQB, and add the idea of constant ratio to the class model. The mystery of the capacity and paint remains open for later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Corresponding points and sides of scaled shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Corresponding sides and points Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the sufurias S, M and L from Lesson 1 and the idea of similar and non-similar sets. In groups of 4 they receive cut-out rectangles A (4 cm by 6 cm), B and C, marked with letters only and with no measurements shown. Without measuring, each learner writes an individual prediction in their notebook: 'Which of B or C is the same shape as A, only bigger? Why do you think so?' Groups then share and record a group prediction and a reason on a half-sheet of paper. Learners also predict: 'If the large sufuria is twice as tall, what do you think we will find when we compare its width to the small sufuria's width?'	Open with: 'Mama Achieng says her three sufurias are the same shape. Last lesson we sorted objects by eye. Today, how can we be sure and not just trust our eyes?' Display the three cut-outs on the board without numbers. Say: 'Silent think time. Write your own prediction first.' WAIT TIME: 15 seconds before anyone speaks. Walk around and read at least 6 notebooks without correcting them. Cold-call 3 groups to share a prediction and reason, asking 'What did you notice that made you choose that one?' Do not say who is correct. Record the three predictions as tally marks on the board (for example, 'B only', 'C only', 'both', 'neither'). Say: 'We have different ideas, so we need evidence. Today we will measure.'	Predict-Observe-Explain (Prediction with Reasoning): committing to a written prediction before measuring makes learners aware of their eyeball judgement, so the later result creates real surprise and a reason to look for a reliable test. Using the sufurias in the same prediction keeps the phenomenon in view.	Check that 100% of learners have a written individual prediction before group talk. Look for reasoning words such as 'looks longer', 'same shape' or 'stretched', which show the preconceptions to challenge (for example, that matching one side length is enough). Note how many groups pick C as similar; this tells you how strong the 'one matching side' idea is. Record names of learners who give no reason, for support in the Observe Phase.
Observe Phase  VIDEO: Corresponding points and sides of scaled shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Corresponding Parts of Similar Figures (Flexbook) Source: CK-12  Search ARES for similar readings	Part 1 (about 25 minutes): Groups measure the sides of rectangles A, B and C to the nearest 0.1 cm using a ruler and check each corner with a set square or the corner of an exercise book. They record results in a table with columns for Rectangle, Length, Width, and Angles. They then calculate ratios of corresponding sides: B to A (length $\frac{6}{4}$, width $\frac{9}{6}$ using the long and short sides in the same order, giving 1.5 for both) and C to A ($\frac{6}{4} = 1.5$ for the short side and $\frac{8}{6} = 1.33$ for the long side). They write down whether the ratios are the same. Part 2 (about 15 minutes): Groups use the measurements of Mama Achieng's sufurias (S: height 12 cm, diameter 16 cm; M: height 18	Distribute rulers, set squares and cut-outs; assign roles: measurer, recorder, checker, reporter, and rotate the roles half way through. Say: 'Measure to the nearest 0.1 cm. The checker must confirm every measurement before it goes in the table.' Circulate and ask: 'Which side of B matches the 4 cm side of A?' and 'What do you get when you divide the long side of C by the long side of A?' For groups that match sides wrongly, say: 'Turn the rectangle so the short side lines up with the short side.' WAIT TIME: 10 seconds after each question before prompting. After the rectangles, ask all groups to hold up a card showing the ratio for the short sides and for the long sides of C. Cold-call 4 groups to	Data Table and Pattern Finding (Analysing and Interpreting Data): organising measurements in a table and calculating ratios side by side makes the constant ratio visible and lets learners see for themselves that one matching ratio is not enough. Comparing rectangles with sufurias links the abstract shape test to the phenomenon.	Look for learners who match corresponding sides correctly (short with short, long with long), measure to the nearest 0.1 cm, and divide image by object consistently. Observable indicator: the group table shows 1.5 and 1.5 for B and 1.5 and about 1.33 for C. Ask a quick check: 'Point to the side of C that does not match the ratio.' Note any group that checks only one pair of sides, or that gets ratios like 0.67 by dividing in the opposite order, and ask them to state the order they used.

	cm, diameter 24 cm; L: height 24 cm, diameter 32 cm), measured directly from real or cardboard models or taken from the data card, and calculate M/S and L/S for height and for diameter: $18/12 = 1.5$, $24/12 = 2$, $24/16 = 1.5$, $32/16 = 2$. They record that the ratios are the same for height and diameter.	report their ratios for B and C. Write the class values on the board: B 1.5 and 1.5, C 1.5 and 1.33. Ask: 'Are the two ratios for C equal? What does that tell you about C?' For Part 2, say: 'Now test Mama Achieng's claim. Is the ratio the same for height and for diameter?' Make sure groups write 'about' when rounding 1.33.		
Explain Phase  VIDEO: Using similarity to estimate ratio between side lengths <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Corresponding Parts of Similar Figures Practice Source: CK-12  Search ARES for similar readings	Learners work in their groups to write a short explanation answering: 'How can we tell whether two figures are similar?' Using their tables, they complete the statement: 'Two figures are similar when all corresponding angles are ____ and all corresponding sides are in the ____ ratio.' They explain why C is not similar to A even though one pair of sides has the ratio 1.5, and why B is similar to A. They then explain why the three sufurias are similar: height ratios and diameter ratios are both 1 : 1.5 : 2. Each learner writes one sentence connecting the result to Mama Achieng: 'Her sufurias are similar because ...'. Groups also try a counter-example: 'Could a rectangle 6 cm by 8 cm ever be made similar to A by changing only one side?'	Say: 'Use your numbers, not your eyes. Who can explain, using evidence, why C is not similar to A?' WAIT TIME: 10-15 seconds, then cold-call 3 learners from different groups. After each answer ask: 'Who can add to that, or disagree with evidence?' Introduce the vocabulary 'corresponding sides', 'constant ratio' and 'similar' only after learners have described the idea in their own words. Write the class definition on the board: 'Similar figures have equal corresponding angles and corresponding sides in the same ratio (a constant ratio).' Make the contrast explicit: 'Same angles alone are not enough here: A, B and C are all rectangles with four right angles, but only B is similar.' Then ask: 'Mama Achieng's large sufuria is exactly twice as tall and twice as wide as the small one. What does that tell us about the ratio?' Confirm: each length of L is 2 times the matching length of S. Do not explain area or capacity; say: 'We have not yet explained why the amount of githeri behaves differently. That is for later lessons. Put it on the board as a question.'	Claim-Evidence-Reasoning (Constructing Explanations and Argument from Evidence): learners state a claim (B is similar, C is not), cite their measured ratios as evidence, and give reasoning (similar figures need a constant ratio for all sides). The rectangle contrast gives the reasoning concrete support before it is applied to the sufurias.	Listen for the key ideas: 'all pairs of sides have the same ratio', 'one ratio is not enough', and 'angles are equal'. Indicator of understanding: a learner explains that C fails because $6/4 = 1.5$ but $8/6$ is about 1.33. Misconception to note: 'C is similar because it is also a rectangle' or 'because one side is 1.5 times'. Use a quick exit statement on a slip: 'True or False: if one pair of sides has ratio 1.5, the figures are similar. Give a reason.' Check that at least 80% write False with a reason.
Driving Question Board (DQB)	Learners return to the Driving Question Board. In groups they write questions on sticky notes or	Say: 'Look at the DQB. Which question from Lesson 1 can we now answer with evidence? Which	Driving Question Board (Tracking Growing Understanding): sorting questions into 'know', 'wonder' and	Check that each group contributes at least 2 questions, and that at least one links the constant ratio of

Creation  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Corresponding Sides of Similar Triangles (PLIX) Source: CK-12  Search ARES for similar readings	paper strips arising from today's work, for example: 'The sufurias have the same ratio for height and diameter, so why does the large one hold much more than twice as much githeri?' and 'Does the same ratio of lengths mean the same ratio of areas?' and 'Why does Mama Achieng's paint cost more than double?' They place the notes on the DQB under the headings 'What we now know', 'What we still wonder' and 'How can we find out?'. One learner per group reads a question aloud. Learners mark which earlier questions from Lesson 1 can now be partly answered, such as 'Are the sufurias really the same shape?'	new questions did today's measuring bring up?' Give 3 minutes of silent writing, then WAIT TIME of 10 seconds before sharing. Cold-call 3 groups to read and place a question. Ask: 'How is this question connected to Mama Achieng's sufurias?' Group similar questions together. Point out the key puzzle: 'All lengths are doubled, but the githeri is not just doubled. What do you think needs to be measured next?' Guide, without answering, to the next step: a way to describe the doubling of lengths and where it comes from, which leads to Lesson 3. Move the Lesson 1 question about the same shape to the 'What we now know' column with the evidence note 'ratio 1 : 1.5 : 2 for height and diameter'.	'find out' lets learners see their own progress and motivates the next investigation, keeping the phenomenon at the centre.	lengths to capacity, paint or price. Quality indicator: questions that refer to evidence from today (for example 'ratios 1.5 and 2') rather than only 'why is it bigger?'. Note learners who only write factual questions and prompt them with 'What would you like to find out about the area or the amount of githeri?'
Model Building Phase  VIDEO: Scale drawings <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Corresponding sides and points Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners open the class model: the growing written and diagram explanation of why capacity and paint do not just double. In groups they add today's idea: 'Similar objects have a constant ratio between all corresponding lengths.' They draw a labelled diagram of sufuria S and sufuria L with height 12 cm and 24 cm and diameter 16 cm and 32 cm, and write the ratios $24/12 = 2$ and $32/16 = 2$ beside the diagram. They also add a short note that rectangles A and B are similar (ratio 1.5) while C is not. Finally they write one 'still to explain' line: 'We know every length of L is 2 times S. We do not yet know what happens to the area and the amount of githeri.' Groups copy the updated model into their notebooks.	Say: 'Our model must show what we are sure about from evidence. What new idea can we add today?' Take ideas from at least 4 learners, then write the agreed wording on the class model chart. WAIT TIME: 10 seconds after asking: 'What would be missing from our model if we stopped here?' Cold-call 2 groups to share how their diagram shows the constant ratio. Check that labels are correct: S height 12 cm, diameter 16 cm; L height 24 cm, diameter 32 cm; M height 18 cm, diameter 24 cm. Remind learners to write 'about' for the 1.33 ratio. Make clear what the model does not yet explain: 'We cannot yet say why L feeds 8 times as many, only that every length is doubled.' Close with: 'Next lesson we will look at how an enlargement from a fixed point produces these similar shapes, and how to find the scale	Cumulative Model Building (Modelling): adding one evidence-based idea each lesson, with a labelled diagram and an explicit 'still to explain' line, makes the model a tool for thinking and shows that constant ratio is necessary but not yet sufficient to explain capacity and paint.	Check each group's model entry for: correct dimensions for S, M and L; ratio 2 for both height and diameter of L to S; a clear statement of the constant ratio idea; and an honest 'still to explain' line. Indicator of strong understanding: a learner can say why a stretched shape (like rectangle C) would not belong in the model. Collect two notebook samples for the next lesson's starter.

		factor.' Do not state k^2 or k^3 or the capacity and paint figures.		
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D. TEACHER REFLECTION

After the lesson, ask yourself: Did every learner commit to a written prediction before measuring? How many groups initially thought C was similar because one ratio was 1.5, and did measurement change their minds with evidence? Which groups matched corresponding sides wrongly, and what helped them correct it? Could learners explain, in their own words, why equal angles alone were not enough? Did the sufuria ratios (1 : 1.5 : 2) feel like a natural extension of the rectangle test or an add-on? Which learners may need extra support with dividing and rounding (for example $8/6 = \text{about } 1.33$)? Did the DQB questions show curiosity about area and capacity, ready for later lessons? What will you change in time use between the two parts of the Observe Phase?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We measured the sides and angles of rectangles A (4 cm by 6 cm), B (6 cm by 9 cm) and C (6 cm by 8 cm). The ratios were 1.5 and 1.5 for B to A, but 1.5 and about 1.33 for C to A. For the sufurias, height ratios (12, 18, 24 cm) and diameter ratios (16, 24, 32 cm) were both 1 : 1.5 : 2.
What did I learn?	Similar figures have equal corresponding angles and all corresponding sides in the same ratio (a constant ratio). If even one pair of sides has a different ratio, the figures are not similar. The three sufurias are similar because every length of M is 1.5 times S and every length of L is 2 times S.
How does this explain the phenomenon?	B is similar to A because both side ratios are 1.5. C is not similar to A because its side ratios (1.5 and about 1.33) are not the same, even though all three are rectangles. Mama Achieng's sufurias are similar because height and diameter both go up in the ratio 1 : 1.5 : 2. We still need to find out why this constant ratio of lengths does not make the githeri and paint just double.

LESSON 3 (80 minutes (2 x 40-minute lessons)): Enlargement and Centre of Enlargement: Tracing Mama Achieng's Sufurias and Triangle ABC

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners find out how an enlargement maps an object onto its image from one fixed point, the centre O. They use measured lengths to find the linear scale factor k, which is the first step in explaining why Mama Achieng's large sufuria is not simply double the small one in every way.
Knowledge	By the end of the lesson, the learner should be able to: (1) determine the centre of enlargement and the linear scale factor of similar figures (KICD SLO b); (2) state that an enlargement maps each point of an object to a point of the image on a straight line through the centre O; (3) calculate $k = \text{image length} / \text{object length} = OA'/OA$; (4) relate the sufuria S to L pair to $k = 2$.
Skills	Tracing an object and its image on plain paper; drawing straight lines through corresponding points to locate O; measuring corresponding sides accurately in cm; calculating ratios and checking that they are constant; communicating reasoning with evidence. NGSS-style practices: Analysing and Interpreting Data, Using Mathematics, Constructing Explanations, Developing and Using Models.
Attitudes	Appreciate the use of similarity and enlargement in real-life situations (KICD SLO h). Responsibility: take care of the objects and instruments used. Unity: share resources and take turns with peers. Core competencies: Creativity and Imagination (drawing images), Critical Thinking and Problem Solving (interpretation and inference). PCI: Environment and technology (use of locally available materials).
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	Lesson 2 showed that corresponding sides of similar figures have a constant ratio, and that the sufuria sizes S:M:L have height and diameter ratios of 1 : 1.5 : 2. Lesson 3 turns that constant ratio into the scale factor k of an enlargement and shows that every pair of corresponding points lines up with one centre O. It establishes that Mama Achieng's S to L pair is an enlargement with $k = 2$, so later lessons can ask what k does to area (Lesson 5) and volume (Lesson 6). Lesson 4 will then draw images from a given O and k.
Safety Notes	Use blunt-tipped pencils and rulers; keep sharp compass points and scissors away from faces. Keep walkways clear around the sufuria models. If real aluminium sufurias are used, check for sharp edges and use cardboard models if unsure. Do not use hot sufurias or food in class. Keep phones and tablets on tables and use them only for taking photos or recording.

B. LESSON OVERVIEW

Learners begin by recalling Mama Achieng's three sufurias at Kibuye Market in Kisumu and predicting how a triangle would look if it were enlarged to double size from a fixed point. In groups of 4 they trace triangle ABC ($AB = 3$ cm, $BC = 4$ cm, $AC = 5$ cm) and its image $A'B'C'$ (6, 8, 10 cm) on plain paper. They join A to A' , B to B' and C to C' , see that the lines meet at one point O, and measure OA, OA' and the sides to find $k = 2$. In the Explain phase they link the triangle to the sufuria S to L pair, where height 12 to 24 cm and diameter 16 to 32 cm also give $k = 2$. The class then adds questions to the DQB, for example why doubling lengths still makes the large sufuria hold so much more, and records the first part of the class model: everything is lengthened by $k = 2$ from a centre, but capacity and paint are not yet explained.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Dilations: center Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall Mama Achieng's sufurias S and L (heights 12 cm and 24 cm, diameters 16 cm and 32 cm) and are shown a small triangle ABC (3, 4, 5 cm) and a single pencil dot labelled O about 6 cm from it. Individually, in 3 minutes, they sketch where they think an image of the triangle would appear if every length were doubled 'from' O, and write a one-line reason. They then compare sketches in their group of 4.	Hold up the small and large sufuria (or cardboard models) and say: 'Mama Achieng's large sufuria is twice as tall and twice as wide as the small one. If we could stretch the small sufuria into the large one, what would we stretch it from?' Give WAIT TIME of 10-15 seconds before taking responses. Then display triangle ABC with point O and say: 'Sketch where the doubled triangle would sit. Do not measure yet; commit to a prediction.' Walk the room and cold-call 3 learners to describe their sketch without saying if it is right. Say: 'Keep your prediction; the evidence will test it.' Record two or three different predictions on the board.	Predict-Observe-Explain (prediction before evidence): committing to a sketch and reason makes learners aware of their own ideas, so when they later trace the real image they can compare what they expected with what the evidence shows. It also surfaces common ideas such as 'the image slides along' or 'the image grows in all directions from the triangle's corner'.	Observe whether sketches show an image of similar shape and roughly double size. Look for learners who place the image touching the triangle, rotated, or unconnected to O. Tally in a notebook how many sketches (a) keep the shape, (b) show O as having any role, and (c) show the image farther from O. Listen for correct use of 'image' and 'object' in group talk. Use this to decide how much support groups need when tracing.
Observe Phase  VIDEO: Dilating triangles: find the error Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: center Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners receive a plain sheet showing triangle ABC (AB = 3 cm, BC = 4 cm, AC = 5 cm, right angle at B) and its enlarged image A'B'C' already drawn on the same sheet, with O hidden. Step 1: measure and record all six sides in a table (object side, image side, ratio image/object). Step 2: with a ruler, draw a straight line through A and A', through B and B' and through C and C', extending them well past the figures. Step 3: mark the point where the three lines meet and label it O. Step 4: measure OA and OA', OB and OB', OC and OC', and calculate OA'/OA, OB'/OB, OC'/OC. Step 5: compare the side ratios (6/3, 8/4, 10/5) with the OA'/OA ratios. Step 6: for the sufuria pair, measure the heights (12 cm and 24 cm) and the diameters (16 cm and 32 cm) on the models or from given cards and calculate image/object. Each learner has a role: measurer, line-	Distribute sheets, rulers and pencils and check that groups share resources fairly: 'Take turns with the ruler; every member must measure at least once.' Model Step 2 once on the board: 'Line the ruler on A and A' and draw the whole line, not just between them.' While groups work, ask: 'What do you notice about where your three lines go?' Give WAIT TIME of 10-15 seconds after each question. If lines miss each other by a few millimetres, say: 'Small measuring error is normal. Do your lines nearly meet at one point?' Cold-call 2 groups to report their side ratios and 2 groups to report OA'/OA. Do not tell them the answer 'k = 2' before the groups have calculated it. Ask groups that finish early: 'Does the answer change if you measure from B instead of A?'	Collecting and organising data in a table, then noticing patterns: learners see that all six ratios are the same and that corresponding-point lines pass through a single point. A structured table helps them move from raw measurements to the pattern, building the idea that an enlargement is fully described by a centre O and a constant ratio k.	Check tables for correct measurements within +/- 0.1 cm and correct ratio calculations (2, 2, 2 for sides; about 2 for OA'/OA, OB'/OB, OC'/OC). Look for lines extended through both points rather than segments only. Use a checklist: (1) lines drawn through all three pairs, (2) O located where lines meet, (3) all ratios recorded, (4) learners say that the ratio is the same for every pair. Note groups whose OA'/OA differs a lot from 2 and prompt remeasurement from O, not from the figure's edge.

	drawer, recorder, checker. They rotate roles for Step 6.			
Explain Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups use their data to write a claim, evidence and reasoning (CER) statement: 'Triangle A'B'C' is an enlargement of triangle ABC with centre O and scale factor $k = \frac{\text{image side}}{\text{object side}}$ because ____.' They then explain how to find the centre of any enlargement (join corresponding points and extend the lines until they meet) and how to find k two ways: image side / object side and OA'/OA. Next they apply the idea to Mama Achieng's pair: they state that S to L has $k = 2$ because $24/12 = 2$ and $32/16 = 2$, and describe the 'centre' as a point from which the small sufuria could be imagined stretched to the large one. In pairs they answer a quick problem: a triangle with sides 3, 4, 5 cm is reduced to sides 1.5, 2, 2.5 cm; what is k? (0.5) and why is it still called an enlargement with a fractional scale factor?	Say: 'Using only your table, tell me what k is and how you know.' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners from different groups. Press for evidence: 'Which numbers in your table show that?' Introduce vocabulary at the point of need: 'The fixed point where all the lines meet is the centre of enlargement, O. The ratio of image length to object length is the linear scale factor, k.' Write $k = \text{image length} / \text{object length} = OA'/OA$. Connect to Lesson 2: 'Last lesson you found a constant ratio of corresponding sides. Today that constant ratio has a name in an enlargement: the scale factor.' Link to the phenomenon: 'Mama Achieng's large sufuria has every length 2 times the small one. So what is k for S to L?' Cold-call 2 learners. Ask: 'If every length is doubled, why might the amount of githeri not just double? Hold that question; we will put it on the Driving Question Board.' Do not explain area or volume yet.	Claim-Evidence-Reasoning with a Think-Pair-Share: learners must support the claim $k = 2$ with their own measured data and connect it to the sufuria dimensions. Sharing in pairs before whole-class talk lets every learner rehearse the mathematical language. Linking to Lesson 2's constant ratio helps them see k as a familiar idea in a new role.	Listen for learners who say 'k is image divided by object' and who can explain that the three lines meet at O because every point is enlarged along a straight line from O. Look for correct use of 'object', 'image' and 'corresponding'. Collect CER statements and sort into three piles: complete (claim, data, reason), missing reasoning, or confusing k with the difference in lengths (e.g. 'image is 3 cm bigger'). Note which learners treat the reduction $k = 0.5$ as 'not an enlargement'. Re-teach with the pair of sufurias if needed: 12 to 24 is times 2, not plus 12.
Driving Question Board (DQB) Creation  VIDEO: Dilations: center Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: center Source: Khan Academy (English	Groups return to the class Driving Question Board: 'Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object?' Each group writes one sticky note with either (a) something we now know, for example 'Lengths of S to L are multiplied by $k = 2$ from a centre', or (b) a new question, for example 'If every length is doubled, what happens to the area of the base?'	Say: 'Everything about the large sufuria's lengths is now explained: $k = 2$. But the puzzle remains. Write what you now know and what you are still wondering.' Give WAIT TIME of 10-15 seconds, then ask each group to post one note. Cold-call 3 groups to read their note aloud and say how it connects to the phenomenon. Group similar questions together: 'Several of you ask about area and about capacity; which lesson will help us?' Point to the lesson list: area in Lesson 5 and volume in	Driving Question Board as a public record of growing understanding: separating 'what we know' from 'what we still wonder' shows learners that the evidence so far (lengths scale by k) answers part of the driving question and clearly points to what to investigate next.	Check that notes link to the phenomenon and are mathematically sensible (for example a 'known' note stating $k = 2$ for S to L). Count how many groups ask about area or capacity, which signals readiness for Lessons 5 and 6. Identify notes that mix up lengths and amounts (for example 'the large sufuria is 2 times bigger so it holds 2 times more') and keep them as testable claims for later lessons without correcting them yet.

- US curriculum) Search ARES for similar readings	or 'Does the centre change the size of the image?'. They place notes in 'Answered so far' or 'Still to investigate' columns and link each to the question.	Lesson 6. Praise questions that are testable: 'How could we find that out?'		
Model Building Phase VIDEO: Dilating triangles: find the error Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Prove triangle similarity <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners open the class model for 'why capacity and paint do not just double'. Individually for 5 minutes, then in groups, they add this lesson's piece: a labelled diagram showing triangle ABC, image A'B'C', centre O, lines through corresponding points, and $k = 2$, next to a sketch of sufuria S and L with heights 12 and 24 cm and diameters 16 and 32 cm and arrows labelled 'x 2'. Under it they write a short sentence: 'Every length of the large sufuria is 2 times the small one, so $k = 2$. We still need to find what this does to area and capacity.' They also fill in the summary table row for this lesson.	Say: 'Your model so far says the sufurias are similar. Add what today's evidence tells us.' Circulate and ask: 'Where is O in your sufuria sketch, and what does $k = 2$ apply to?' Give WAIT TIME of 10-15 seconds. Cold-call 2 groups to show their diagram. Check that the model does not claim capacity is 8 times or paint 4 times yet. Ask: 'What part of the puzzle does $k = 2$ explain, and what part is still missing?' Close with: 'Next lesson you will draw images yourself from a centre O with a given k.'	Cumulative model revision: adding one evidence-based piece at a time keeps the model honest. Learners must decide what the new data supports (lengths scale by k) and what it does not yet support (area and capacity), which models scientific and mathematical habits of mind.	Look for models that correctly show O, the lines through corresponding points, $k = 2$ with the working $6/3 = 8/4 = 10/5$, and the sufuria heights and diameters with x 2. Flag any model that states a capacity or area result prematurely. Ask each learner to complete the exit question: 'How do you find the centre of enlargement and k from an object and its image?' Success means mentioning joining corresponding points to find O and dividing an image length by the matching object length for k.

D. TEACHER REFLECTION

Which learners were able to find O by joining corresponding points and which needed help extending the lines? Did groups confuse k with the difference in lengths (adding 3 cm instead of multiplying by 2)? Were any lines not meeting at one point because of poor measuring, and how did I use that to discuss measurement error? Did learners connect triangle ABC to the sufuria S to L pair, or did they treat them as separate tasks? Which DQB questions about area and volume should be used to open Lessons 5 and 6? What should I adjust before Lesson 4, in which learners draw images from a given O and k?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We traced triangle ABC (3, 4, 5 cm) and its image A'B'C' (6, 8, 10 cm), joined A to A', B to B' and C to C', and saw that the lines met at one point, O. All side ratios and OA'/OA , OB'/OB , OC'/OC were 2. The sufuria heights (12 to 24 cm) and diameters (16 to 32 cm) also doubled.
What did I learn?	An enlargement maps each point of an object to the image along a straight line from a fixed centre O. The centre is found by joining corresponding points. The linear scale factor $k = \text{image length} / \text{object length} = OA'/OA$. For Mama Achieng's sufurias S to L, $k = 2$.

How does this explain the phenomenon?

The large sufuria is an enlargement of the small one with $k = 2$ because every length (height and diameter) is multiplied by 2. This explains why the large sufuria looks like a doubled copy, but we still need to find out what $k = 2$ does to the paint area and the capacity.

LESSON 4 (80 minutes (2 x 40-minute lessons)): Drawing the Image: Enlarging and Shrinking Triangle ABC from Centre O

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners draw the image of an object given a centre of enlargement O and a positive linear scale factor ($k > 1$ and $0 < k < 1$) on plain and squared paper, check the image by measurement and digital zoom, and see how drawing the sufuria-style enlargement prepares them to ask how area and volume change.
Knowledge	By the end of the lesson the learner should be able to: c) draw the image of an object given the centre of enlargement and the linear scale factor. The image point A' lies on the ray from O through A with $OA' = k \times OA$. Triangle ABC ($AB = 3$ cm, $BC = 4$ cm, $AC = 5$ cm) enlarged by $k = 2$ gives sides 6, 8, 10 cm; by $k = 0.5$ gives 1.5, 2, 2.5 cm. Angles stay equal and sides stay in the constant ratio k (builds on Lesson 3: centre O, $k = OA'/OA$, corresponding-point lines).
Skills	Drawing with ruler and set square; measuring lengths to the nearest millimetre; calculating image distances ($OA' = k \times OA$); counting squares from a centre on squared paper (rectangle A, 4 cm by 6 cm); using a phone or tablet to zoom and shrink an image and compare; checking and correcting work. NGSS-style practices: Developing and Using Models (draw an image as a model), Analyzing and Interpreting Data (compare measured and predicted lengths), Constructing Explanations.
Attitudes	Core competencies: Digital literacy (using zoom and shrink tools on a device); Creativity and Imagination (drawing images under a given scale factor). Value: Responsibility (taking care of rulers, devices and the sufuria models). PCI: Environment and technology. Appreciate the use of similarity and enlargement in real-life situations such as Mama Achieng's three sufurias.
Key Inquiry Question	How is similarity and enlargement applied in real life? Focus question: If we know the centre O and k, how can we draw an exact image, and what stays the same and what changes when we draw it?
Purpose in Storyline	Lesson 3 taught learners to find O and k from an object and its image. Lesson 4 reverses it: given O and k, learners build the image themselves. This gives learners reliable enlarged and reduced figures (triangle ABC, rectangle A) for Lesson 5, where they compare areas (k^2), and Lesson 6, where they compare volumes (k^3), to explain why Mama Achieng's large sufuria ($k = 2$) holds far more than twice the githeri. This lesson only draws and checks lengths and angles; it does not yet explain the area or volume change.
Safety Notes	Use blunt-tipped pencils and compasses; do not pass sharp points between learners. Keep rulers flat on desks and away from faces. Keep devices on desks and charge them beforehand; if no device is available, use the teacher's single device for a demonstration. Handle cardboard or aluminium sufuria models carefully (no sharp edges). Keep the offline zoom activity within the school's online-safety rules: use only the pre-saved image file, no internet browsing.

B. LESSON OVERVIEW

Learners start by predicting where the image of a point will fall when Mama Achieng's small sufuria (S) is enlarged by $k = 2$ from a centre O, then draw the image of triangle ABC (3, 4, 5 cm) on plain paper using rays from O, for $k = 2$ (sides 6, 8, 10 cm) and $k = 0.5$ (sides 1.5, 2, 2.5 cm). They then draw rectangle A (4 cm by 6 cm) on squared paper with the same method by counting squares, check each image by measuring sides and angles, and use a phone or tablet to zoom and shrink a photo of a sufuria to compare with their drawings. In the explain phase they state a rule for drawing any image, and they add the drawing method to the DQB and the class model, noting that all lengths double but they still need to find out what happens to area and volume.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilating shapes: shrinking Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations and properties Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners sit in groups of 4 with plain paper and rulers. The teacher shows a dot O on the board and a point A that is 3 cm from O, and says Mama Achieng's small sufuria is enlarged by $k = 2$ to make the large one. Individually, for 2 minutes, learners sketch and write where A' will be if $k = 2$ and where it will be if $k = 0.5$, and predict the lengths of the sides of triangle ABC (3, 4, 5 cm) after each enlargement. They also predict: Will the angle at B change? Will the shape stay the same? They compare in their group and mark each prediction as agree or disagree.	Draw O and A on the board. Say: 'Mama Achieng's large sufuria is 2 times as tall and 2 times as wide as the small one. Last lesson we found O and k from an object and its image. Today we are given O and k. Where do you think the image point A' will be? Sketch it and write one reason.' Give WAIT TIME of 10-15 seconds before anyone speaks, then 2 minutes of silent writing. Cold-call 4 learners from different groups to share (not volunteers): 'Which direction from O? How far?' Record all predictions on the board without judging. Ask: 'What did you use from Lesson 3 to decide?' (expected: $k = OA'/OA$ and the line through O and A). Mark one correct and one wrong prediction for later comparison.	Predict-Share-Revisit (Think-Pair-Share): committing to a written prediction before drawing makes learners reveal their idea of the rule linking O, A and A', so later measurement can confirm or challenge it. The strategy builds understanding by creating a cognitive gap that the drawing activity will fill.	Observe: (1) Do learners place A' on the line from O through A? (2) Do they use the scale factor to decide the distance (6 cm for $k = 2$, 1.5 cm for $k = 0.5$)? (3) Do they claim the angle at B will change? Tally on a clipboard how many groups placed A' correctly for $k = 2$ and for $k = 0.5$. Misconception to note: adding 2 cm instead of multiplying by 2, or placing A' beside A instead of on the ray from O.
Observe Phase  VIDEO: Dilating triangles: find the error Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations and properties Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Task 1 (plain paper, $k = 2$): Learners mark O near the left of the page and draw triangle ABC ($AB = 3$ cm, $BC = 4$ cm, right angle at B, $AC = 5$ cm) about 4 cm from O. They draw faint rays from O through A, B and C, measure OA, OB and OC, calculate $OA' = 2 \times OA$, $OB' = 2 \times OB$ and $OC' = 2 \times OC$, mark A', B', C' on the rays, join them, then measure A'B', B'C' and A'C' (expect 6, 8, 10 cm) and check the right angle at B' with a set square. Task 2 ($k = 0.5$): they repeat with the same triangle and centre, marking points halfway from O (expect 1.5, 2, 2.5 cm). Task 3 (squared paper): they draw rectangle A (4 cm by 6 cm) with one corner 2 squares from O, use $k = 2$ by counting squares along	Model the first point only: 'Watch me. I measure OA, it is 4 cm. For $k = 2$ I need $OA' = 2 \times 4 = 8$ cm. I measure 8 cm along the ray from O through A and mark A'.' Then release groups. Circulate with a checklist; ask: 'How do you know A' is on the ray? Where does the ray go if you extend it backwards?' Give WAIT TIME of 10-15 seconds after each question. Pause the class after Task 1 and cold-call 3 groups for their measured A'B', B'C', A'C' and write them in a class table next to predictions (6, 8, 10 cm). Ask: 'Which lengths changed? Which stayed the same?' For the digital zoom, assign roles (device holder, measurer, recorder, checker) so every learner takes part; remind:	Data table with measured vs predicted values (Analyzing and Interpreting Data): learners record object length, image length and ratio for every side, then notice that every ratio equals k while the angles remain equal. Drawing the same figure for $k = 2$ and $k = 0.5$ lets them see enlargement and reduction as one rule, and comparing paper drawing, squared-paper counting and digital zoom is triangulation of evidence.	Observable indicators: (1) A', B', C' lie on rays through O, A, B, C (check with a ruler laid across); (2) image sides within 2 mm of 6, 8, 10 cm ($k = 2$) and 1.5, 2, 2.5 cm ($k = 0.5$); (3) learners check the right angle at B' with a set square; (4) on squared paper, the rectangle image is 8 cm by 12 cm; (5) learners write ratio image/object = k in each table row. Use a quick rating scale: Exceeds (draws both images accurately and explains a check), Meets (draws both with minor errors), Approaches (draws one correctly), Below (needs guidance). Common errors to note: measuring from A instead of from O; adding k instead of multiplying; mixing up object and image.

	each ray to draw the image (8 cm by 12 cm), and record in a table the object length, image length and ratio. Task 4 (digital zoom): using a phone or tablet with a saved photo of a sufuria, learners zoom in to double the width and then shrink to half, measure the screen width with a ruler, and compare with the drawings.	'Take care of the devices and use only the saved image.' For learners who finish early, ask them to draw the image with $k = 3$ or $k = 1.5$ and predict the lengths first. Support learners who struggle by giving a partly completed ray diagram. Watch for measurement errors and ask groups to re-measure rather than telling them the answer.		
Explain Phase  VIDEO: Dilating shapes: shrinking by 1/2 Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups compare their drawings and measurements and write a short rule titled 'How to draw the image given O and k'. They explain, in their own words and with a labelled sketch, why $OA' = k \times OA$, why the image points must be on the rays through O and the object points, and why the shape stays the same while lengths change by the factor k. They discuss what happens to the position of the image when $k > 1$ (image is further from O and bigger) and when $0 < k < 1$ (image is nearer to O and smaller). They connect this to Mama Achieng's small and large sufurias: all lengths of the large sufuria (height 24 cm, diameter 32 cm) are 2 times those of the small one (height 12 cm, diameter 16 cm). Each group answers: 'We drew a bigger and smaller triangle, but did the angles change? Did the shape change? What would the area be? We have not measured yet.'	Say: 'Using your tables, tell your group in one sentence what you did each time to find a new point. Then write the steps in order.' Give WAIT TIME of 10-15 seconds before group talk starts. Cold-call 4 groups to read one step each, building a class list on the board: (1) mark O, (2) draw rays through each object point, (3) measure OA etc., (4) multiply by k to get OA', (5) mark and join, (6) check by measuring. Ask probing questions: 'Why can't we just add 3 cm to each side?' 'What would happen to the image if $k = 1$?' 'Mama Achieng's large sufuria has height 24 cm. What k links it to the small one with height 12 cm?' Then ask: 'Is the large sufuria 2 times as big in every way? Hold that question for the next lessons.' Correct any claim that area also doubles by saying: 'Good thought, we will test that with evidence in Lesson 5. For now we only checked lengths.' Do not teach k^2 or k^3 here.	Claim-Evidence-Reasoning with a written procedure: groups state the claim (the rule for drawing the image), cite evidence (their measured data and the digital zoom check) and give reasoning ($k = OA'/OA$ from Lesson 3 reversed). Writing the steps in order turns a drawing activity into a transferable method, supporting Constructing Explanations.	Listen for: correct use of vocabulary (object, image, centre, scale factor, ray); the statement ' $OA' = k \times OA$ '; explanation that angles stay equal and sides are in the same ratio k; correct distinction between $k > 1$ and $0 < k < 1$. Collect one exit slip per group with the steps. Red flags: claiming the image is congruent to the object, or saying area also doubles without evidence. Ask 3 learners to explain to a partner and check that all three can state the rule.
Driving Question Board (DQB) Creation  VIDEO: Dilating shapes: shrinking Source: Khan	Learners go to the DQB and add sticky notes: what they now know (e.g., 'To draw the image we multiply the distance from O by k'), what they confirmed from the previous lesson, and new questions, such as 'The sufuria is 2 times as tall and wide, but how	Say: 'Write one thing you can now do, and one question about the sufuria that drawing has raised. One idea per note.' Give 1 minute of WAIT TIME for writing. Invite 5 learners (cold-call, from different groups) to place and read their notes aloud. Ask: 'Which of these	Driving Question Board (question tracking): sorting new and answered questions makes progress visible and keeps learners' own wonderings steering the next lessons. Grouping questions by theme helps learners see how drawing images connects	Check that every group has contributed a note, that notes use mathematical language correctly, and that at least some questions are about area or volume (showing that learners link the drawings to the phenomenon). Note learners who write only what they did rather

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: center <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	much more paint covers the outside?' and 'If we draw the image of a rectangle with $k = 3$, how many times bigger is the space inside?' They place each note under a heading (Drawing and k, Checking, Area and Volume Questions). They tick or move questions that have been answered, for example 'Where does the image go on the page?', and mark those that remain open (capacity, paint cost, servings).	questions can the drawings we made today help to answer? Which need more evidence?' Cluster questions about area and volume and label them 'Lessons 5 and 6'. Move answered questions to the 'We found out' column. Link back to the driving question: 'Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri?' Ensure that every group has at least one note on the board.	to the later work on area and volume.	than a question. Use a simple tally of the number of questions per theme to plan Lesson 5.
Model Building Phase VIDEO: Dilating triangles: find the error Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Use similar triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Groups revise the class model (the written and diagram explanation of why capacity and paint do not just double). They add a labelled diagram showing triangle ABC and its images for $k = 2$ and $k = 0.5$ from the same centre O, with the lengths, and a sentence such as 'Every length of the large sufuria is $k = 2$ times the small one, as when we drew triangle A'B'C' with sides 6, 8, 10 cm from 3, 4, 5 cm.' They complete the summary table below, and write what the model cannot yet explain: how much the area (paint) and the volume (githeri) increase. Each group writes one prediction to test in Lesson 5, such as 'If lengths double, the area of the rectangle image (8 by 12) will be ___ times the object (4 by 6).'	Say: 'Our model so far says the large sufuria is a $k = 2$ enlargement of the small one. What new evidence from today can you add? What can we still not explain?' Provide WAIT TIME of 10-15 seconds, then 3 minutes for groups to add to the model. Cold-call 3 groups to share their revision and point to the specific piece of evidence (a measured side or the squared-paper table). Ask: 'Does this model yet explain why customers say the large sufuria feeds far more than twice as many people?' (expected: No, because we have only looked at lengths.) Highlight that the price KSh 120 vs KSh 60 follows the linear factor, but do not explain area or volume. Give the next-lesson hook: 'Next time we will compare the areas of our rectangle and triangle images with the objects. Make a prediction: when lengths double, will the area double?' Collect the group models.	Cumulative model revision: groups add today's evidence to a shared explanatory model and mark the gaps. Distinguishing 'what the model explains' from 'what it cannot yet explain' strengthens epistemic thinking and prepares for the area and volume scale factor lessons.	Check that each model (1) shows both $k = 2$ and $k = 0.5$ images with correct lengths (6, 8, 10 cm and 1.5, 2, 2.5 cm), (2) states that angles are unchanged and sides are in ratio k, (3) links to the sufuria as a $k = 2$ enlargement without claiming that capacity or paint simply double, and (4) identifies at least one gap. Provide feedback as 'one star and one wish' on each model. Use a short exit question: 'State the distance OA' if OA = 5 cm and $k = 0.5$.'

D. TEACHER REFLECTION

After the lesson, reflect: Which learners placed image points on the ray from O and used multiplication rather than addition by the end? Which groups measured from O

correctly, and which measured from the object point? How accurate were the drawings (within 2 mm) on plain versus squared paper? Did the digital zoom comparison help learners see that the shape stays the same, or did screen size confuse them? Did any learners claim that area doubles when lengths double, and how will I use that as a prediction in Lesson 5? Was there enough time for all four tasks, or should Task 4 move to the start of the next lesson? Which learners need scaffolded diagrams or extension tasks ($k = 3$, $k = 1.5$)? How well did learners with limited access to devices participate?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We drew triangle ABC (3, 4, 5 cm) from centre O with $k = 2$ and got an image with sides 6, 8, 10 cm; with $k = 0.5$ we got sides 1.5, 2, 2.5 cm. On squared paper, rectangle A (4 cm by 6 cm) with $k = 2$ gave an image 8 cm by 12 cm. The angles stayed equal and the digital zoom matched our drawings.
What did I learn?	To draw the image, draw a ray from O through each object point, measure OA, calculate $OA' = k \times OA$, mark A' on the ray and join the image points. When $k > 1$ the image is bigger and further from O; when $0 < k < 1$ it is smaller and nearer to O. Each image side is k times the object side and the shape stays similar.
How does this explain the phenomenon?	Mama Achieng's large sufuria is a $k = 2$ enlargement of the small one, so every length (height 12 to 24 cm, diameter 16 to 32 cm) is doubled, just like our triangle sides. This explains why it looks the same shape and why the KSh 120 price follows the doubled lengths. It does not yet explain why capacity and paint do not just double, which we will investigate with area (Lesson 5) and volume (Lesson 6).

LESSON 5 (80 minutes (2 x 40-minute lessons)): Area Scale Factor: Why Paint Costs Rise Faster Than Price

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners find the area scale factor by comparing the area of an image with the area of its object, discover that it equals the square of the linear scale factor (k^2), and use it to explain why painting the large sufuria costs about 4 times more than the small one.
Knowledge	By the end of the lesson the learner should be able to: determine the area scale factor of similar figures; relate the linear scale factor to the area scale factor. Area scale factor = image area / object area = k^2 . A 3 cm by 2 cm rectangle (area 6 cm ²) enlarged with $k = 3$ gives 9 cm by 6 cm (area 54 cm ²), and $54/6 = 9 = 3^2$. Outside paint area uses two lengths (radius and height), so it multiplies by k^2 .
Skills	Calculating areas of rectangles and sufuria outside surfaces ($A = 2\pi r h + \pi r^2$, $\pi = 3.14$); calculating ratios of image area to object area; using squared paper to count unit squares; analysing and interpreting data (NGSS SEP: analysing data to find patterns); developing and revising a model (NGSS SEP); using mathematics to make and test predictions; communicating reasoning with peers (collaboration).
Attitudes	Appreciate the use of similarity and enlargement in real-life situations such as costing paint for containers. Responsibility: take care of the rulers, paper and sufuria models. Respect: listen to peers' ideas while discussing patterns. Critical thinking and problem solving: build interpretation and inference skills while testing a pattern on new data.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	Lessons 1-4 established that S, M and L are similar, that corresponding lengths are in a constant ratio, and that an enlargement from a centre O with scale factor k draws an image. Lesson 5 is the first step in explaining why 'everything doubled' does not mean 'paint doubled': learners discover that area changes by k^2 , which explains the paint cost (KSh 20 to KSh 80, four times). The model is revised to explain paint and surface. The puzzle of capacity (k^3) is left open for Lesson 6.
Safety Notes	Use blunt-tipped scissors and rulers under supervision. Keep bags and sufuria models off walkways. Real sufurias must be clean and cool; use cardboard models where possible. Learners should not handle oil paint in class; paint costs are worked from given data only. Use learner phones or tablets only for teacher-approved tasks and follow online safety rules.

B. LESSON OVERVIEW

Mama Achieng at Kibuye Market in Kisumu paints the outside of her small (S), medium (M) and large (L) sufurias with oil paint. The large one is 2 times as tall and 2 times as wide as the small one, yet its paint costs about KSh 80 against KSh 20 for the small one, which is 4 times and not 2 times. In this lesson learners predict what happens to area when lengths are enlarged, then gather evidence in two steps. First they enlarge a 3 cm by 2 cm rectangle with $k = 3$ on squared paper and compare the areas (6 cm² to 54 cm², factor 9). Then they work out the outside area of S, M and L (803.84, 1808.64 and 3215.36 cm²) and the ratios to S (1, 2.25 and 4). Learners explain the pattern as area scale factor = k^2 because area depends on two lengths. They add questions to the DQB and revise the class model to explain paint and surface. Capacity (volume) is deliberately left for Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	(About 10 minutes) Learners recall	Hold up the 3 cm by 2 cm	Prediction with a confidence rating	Look for: every learner has a

 VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Mama Achieng's three sufurias and the facts from earlier lessons: S, M and L are similar, and L has $k = 2$ from S. The teacher shows the cut-out 3 cm by 2 cm rectangle and says it will be enlarged with $k = 3$. Individually, then in groups of 4, learners predict on a Predict card: (1) the new length and width, (2) the new area, and (3) whether the paint for the large sufuria will cost 2 times, 4 times or 8 times the small one, with a reason. They sketch their prediction as a diagram or a number sentence, for example 'area $\times 3$' or 'area $\times 9$'. They record the prediction in their books with a confidence rating from 1 to 5.	rectangle and say: 'Mama Achieng's large sufuria is 2 times as tall and 2 times as wide as the small one, but her paint for it costs KSh 80 and not KSh 40. Today we find out why. Before I teach anything, I want your best prediction.' Pose: 'If I enlarge this rectangle with $k = 3$, what will its area be?' Give individual silent think time: WAIT TIME 15 seconds, then ask learners to write the prediction. Circulate and note the three common ideas ($\times 3$, $\times 6$, $\times 9$). Cold-call 4 groups to share a prediction and reason without judging: 'Which group will tell us what number they expect for the new area, and why?' Record all predictions on the board in tally form. Say: 'Keep your prediction card, we will test it with evidence.' Do not confirm or correct at this point. Check that all learners, including quiet ones, have written something.	and Public Prediction Tally. Writing a prediction exposes the common intuition that 'area multiplies by k'. Learners will later compare it to evidence, which creates productive cognitive conflict and ownership of the idea.	written prediction with a reason; number of learners predicting $\times 3$, $\times 6$ or $\times 9$ on the tally; whether learners use the earlier language 'k' and 'linear scale factor' correctly. Indicator of prior-knowledge gap: a learner who calculates the new sides (9 cm by 6 cm) correctly but still says the area is $6 \times 3 = 18 \text{ cm}^2$.
Observe Phase  VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(About 30 minutes) PART A: Rectangle enlargement. In groups of 4 with squared paper, rulers and scissors, learners draw the 3 cm by 2 cm rectangle (object) and its image with $k = 3$ (9 cm by 6 cm) using the drawing skills from Lessons 3-4. They count the unit squares (1 cm^2) in each and complete a table: object area 6 cm^2, image area 54 cm^2, image area / object area = 9, $k = 3$, $k^2 = 9$. Learners cut the object into 1 cm squares and try to tile the image with copies of the object, finding that exactly 9 copies fit (3 along, 3 across). Groups also try $k = 2$ (6 cm by 4 cm, area 24 cm^2, ratio 4) and $k = 0.5$ (1.5 cm by 1 cm, area 1.5 cm^2, ratio 0.25) to test the pattern. PART B: The	Introduce the task: 'You are paint inspectors for Mama Achieng. Your job is to find out how the area changes when the lengths are enlarged.' Distribute materials; assign roles (measurer, drawer, counter, recorder) and rotate roles between Part A and Part B. Move around and ask: 'How many of the small rectangles fit into the big one? Show me with your cut-outs.' and 'Why does 3 along and 3 across give 9 and not 3?' Use WAIT TIME of 10-15 seconds after each question. For groups that struggle, prompt: 'What is 9×6? What is 3×2? How do these compare?' Check arithmetic in Part B by cold-calling 3 groups for the value of the outside area of L (3215.36) and asking how they got	Structured data collection and Pattern Seeking using tables. Learners gather image and object areas from physical tiling and calculation, see the same pattern across different k values and across a real and a mathematical context, and generate their own 'We noticed...' statements. Physical tiling gives a concrete picture of why the area factor is a square.	Observable indicators: correct unit-square counts (6 and 54); correct area ratio of 9 for $k = 3$; correct outside areas for S, M and L; the table is completed with ratios 2.25 and 4; learners tile the image with 9 copies; the group's 'We noticed...' statement links the area ratio to k in some form. Teacher uses a checklist: drawing accuracy, calculation accuracy, and quality of the noticing statement. Common errors to log: counting only the border, using $2 \times k$ for area, or missing the base in the sufuria area.

	sufuria paint area. Using the data card, learners work out the outside area to be painted (side + base, $A = 2\pi r h + \pi r^2$, $\pi = 3.14$): $S = 602.88 + 200.96 = 803.84 \text{ cm}^2$; $M = 1356.48 + 452.16 = 1808.64 \text{ cm}^2$; $L = 2411.52 + 803.84 = 3215.36 \text{ cm}^2$. They divide each area by the area of S: M gives 2.25 and L gives 4. They link these to the paint cost at about KSh 1 per 40 cm^2: S about KSh 20, M about KSh 45, L about KSh 80. Each group completes a table with columns k, k^2, area ratio to S and paint cost ratio.	it. Intervene on common errors: forgetting the base (the sufuria is open at the top, so there is no lid area); using diameter instead of radius. Ask groups to check the pattern: 'Does the ratio for L match $k = 2$ squared?' Ask one group to report a pattern in each table. Do NOT state the rule k^2 yet; ask groups to write 'We noticed...' statements. Remind learners about safety with scissors and care with materials.		
Explain Phase  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(About 25 minutes) Groups share 'We noticed...' statements in a gallery walk, then the class builds a consensus claim: 'When the length is enlarged by k, the area is multiplied by k^2, so area scale factor = image area / object area = k^2.' Learners use the Claim-Evidence-Reasoning (CER) frame to explain the sufuria paint puzzle: Claim: painting L costs about 4 times as much as S. Evidence: the outside area of L is 3215.36 cm^2 against 803.84 cm^2 for S, ratio 4; paint cost about KSh 80 against KSh 20. Reasoning: area depends on two lengths (for the rectangle, length and width; for the sufuria, radius and height), and each is multiplied by $k = 2$, so the area is multiplied by $2 \times 2 = 4$. They also explain why the price of KSh 120 for L (only 2 times KSh 60) means Mama Achieng's paint costs grow faster than her price. They answer: What would the area factor be for M ($k = 1.5$)? (2.25, which matches the data.) Then they predict: for $k = 5$, the area factor is 25. Finally, they return to their original prediction cards and mark what	Run a gallery walk: 'Visit two other groups. Find one statement that is the same as yours and one that is different.' Then call the class together and ask 5 cold-calls: 'What did group X notice about the area ratio and k?' Build the shared claim on the board using learners' words and add the symbols k^2. Ask: 'Why a square? Think back to the 3 along and 3 across tiling. Turn and talk with your partner.' WAIT TIME 15 seconds, then cold-call 3 pairs. Provide the CER frame on the board and have groups write a CER paragraph for the paint puzzle. Ask: 'What would be the area factor for M, $k = 1.5$? Check it against your table.' (2.25.) Press for precision: 'Is the area factor 2 or 4 for the large sufuria? Which length numbers did you multiply?' Use a misconception probe: 'Joseph says that because the large sufuria is twice as big, the paint is doubled. What evidence would you show Joseph?' Then name the idea explicitly: 'Area scale factor equals k squared.' Note that the capacity question (how much githeri) is for	Claim-Evidence-Reasoning (CER) with Turn-and-Talk and consensus-building. CER forces learners to connect the data to the mathematical rule and to the phenomenon; peer talk makes reasoning visible; revisiting the predictions supports conceptual change.	Indicators: learners state area scale factor = k^2 and can predict the factor for a new k (for example $k = 5$ gives 25, $k = 0.5$ gives 0.25); the CER paragraph includes data (803.84, 3215.36, ratio 4), the rule, and the reason (two lengths); learners can say why the price (KSh 120) does not match the paint cost (KSh 80). Teacher listens for the misconception 'area doubles when length doubles' and checks that learners now reject it with evidence. Exit ticket (3 minutes): A rectangle is enlarged with $k = 4$. What is the area scale factor? If the object area is 5 cm^2, what is the image area? (16; 80 cm^2).

	they got right and what they would change.	Lesson 6; ask learners to note that capacity uses three lengths but do not teach it. Return to the Predict cards: 'Which prediction was close? What changed in your thinking?'		
Driving Question Board (DQB) Creation  VIDEO: Dilating shapes: shrinking <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface area <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(About 5 minutes) Learners return to the Driving Question Board with the question: 'Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object?' On sticky notes, groups (a) add answers they now have, for example 'Paint area for L is 4 times that of S because area scale factor = k^2 ', and (b) add new questions, for example 'Does capacity multiply by 4 too, or by more?', 'Does k^2 work for a triangle or a circle?', 'What if k is less than 1?', 'How many litres of githeri will the large sufuria hold?'. They sort notes into 'answered', 'partly answered' and 'still to investigate', and mark the 'still to investigate' questions that Lesson 6 will tackle.	Point to the DQB and say: 'Which of our questions have we moved forward today? Come and move a sticky note and tell us why.' Invite 3 learners to place answered notes with evidence. Then ask: 'What new questions are you left with? Write one on a note.' WAIT TIME 10 seconds before collecting. Cold-call 3 groups to read a new question aloud. Highlight the capacity questions: 'The paint is 4 times, but customers said the githeri feeds far more than twice as many people. What do you think the capacity factor will be? Keep that question for next lesson.' Do not answer the capacity question now. Make sure the quieter learners' questions are included and thank every contribution.	Driving Question Board as a public record of shifting understanding. Sorting notes into answered / partly answered / to investigate helps learners see progress and sets a purpose for the next lesson, and student-generated questions sustain motivation.	Indicators: every group adds at least one answer note and one new question; learners' answers use k^2 correctly; new questions show curiosity about volume or other shapes (evidence of transfer); at least one group connects to capacity and k^3 without being told. Teacher records which questions to carry into Lesson 6.
Model Building Phase  VIDEO: Dilating shapes: shrinking by 1/2 <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE:	(About 10 minutes) Groups revise the class model of the sufuria phenomenon, which so far explained that S, M and L are similar with constant ratio and enlargement by k. They add a new 'Area' layer: a diagram showing the small sufuria (outside area 803.84 cm ²) beside the large one (3215.36 cm ²) with $k = 2$ labelled on the height and diameter, and a box showing 'area $\times k^2 = x \times 4$ '. Next to it they write the paint costs (KSh 20, 45, 80) and a one-sentence explanation: 'Paint area	Hand out the previous model sheet (or the class chart) and say: 'Our model must now explain the paint. Add what we know and mark what we still do not know.' Circulate and ask: 'Where on your diagram can I see k? Where can I see k squared? Show me how your model explains KSh 80 against KSh 20.' WAIT TIME 10-15 seconds after each question. Cold-call 3 groups to explain their revised model in 30 seconds each, and ask the class to give one 'star' (what is clear) and one 'wish' (what	Cumulative Model Revision. Learners add a new component (area, k^2) to an existing explanatory model, link it to the data for the phenomenon (paint cost), and mark the gap (capacity). Revising rather than restarting shows how knowledge builds over the sequence.	Indicators: the revised model shows $k = 2$ and area $\times 4$; the numbers 803.84 and 3215.36 (or ratio 4) and KSh 20 and KSh 80 appear; the written explanation states that paint area depends on two lengths so it multiplies by k^2 ; the capacity box is left as an open question rather than guessed as $\times 4$ with no reason. Use a 4-point rubric (missing, partial, secure, extended) and note which groups still confuse k and k^2 .

Compare areas by multiplying <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	multiplies by k^2 , so the large sufuria needs about 4 times the paint, not 2 times.' They also add a placeholder 'Capacity: ? (Lesson 6)' with their current prediction. Learners record in the summary table what they observed, what they learned, and how they can explain the phenomenon better now.	is missing). Praise labelling of units (cm ² , KSh). Close by connecting to the driving question: 'We can now predict how length and area change. Next lesson we find out how volume changes, and why the large sufuria feeds far more people than twice as many.' Collect model sheets for later lessons and store the models safely.		
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D. TEACHER REFLECTION

After the lesson, reflect: (1) How many learners moved from predicting 'area $\times k$ ' to 'area $\times k^2$ ' and what evidence changed their minds? (2) Did the tiling activity help learners see why the factor is a square, or did they only copy the numbers? (3) Which groups made errors in the sufuria area calculation (forgetting the base, using diameter for radius), and do I need to reteach $A = 2\pi r h + \pi r^2$ with $\pi = 3.14$? (4) Did learners keep k and k^2 separate when explaining the price (follows k) and the paint (follows k^2)? (5) Were quieter learners heard in cold-calls and DQB notes? (6) Which DQB questions about capacity should open Lesson 6, and which learners are already guessing k^3 ? Plan: group learners who still confuse k and k^2 for a short re-tiling activity with $k = 2$ before Lesson 6, and prepare cubes of side 5 cm and 10 cm for next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We enlarged a 3 cm by 2 cm rectangle (area 6 cm ²) with $k = 3$ to 9 cm by 6 cm (area 54 cm ²), so the area ratio was 9. We worked out the outside areas of the sufurias: S 803.84 cm ² , M 1808.64 cm ² , L 3215.36 cm ² , with ratios to S of 1, 2.25 and 4, and paint costs of about KSh 20, KSh 45 and KSh 80.
What did I learn?	The area scale factor is image area divided by object area, and it equals k^2 . Doubling the lengths ($k = 2$) multiplies the area by 4, and tripling ($k = 3$) multiplies it by 9. Area depends on two lengths, so both are multiplied by k . The price of KSh 120 follows only k , but the paint follows k^2 .
How does this explain the phenomenon?	Mama Achieng's large sufuria is 2 times as tall and 2 times as wide as the small one, so its outside area is $2^2 = 4$ times larger. That is why painting it costs about KSh 80 against KSh 20, which is more than double. We still need to find out why the githeri capacity grows by even more (Lesson 6).

LESSON 6 (80 minutes (2 x 40-minute lessons)): Volume Scale Factor: Why the Large Sufuria Holds 8 Times the Githeri

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners find, from measured volumes of cubes and cuboids, that the volume scale factor of similar solids is k^3 , and use it to explain why Mama Achieng's large sufuria holds 8 times (not 2 times) the githeri of the small one.
Knowledge	By the end of the lesson the learner should be able to: (e) determine the volume scale factor of similar objects, and (f) relate linear scale factor to area and volume scale factor of similar objects. Volume scale factor = k^3 where k is the linear scale factor. Capacity depends on three lengths, so it multiplies by $k \times k \times k$.
Skills	Measuring and calculating the volumes of cubes and cuboids; comparing image volume to object volume as a ratio; using the pattern k , k^2 , k^3 across Lessons 5 and 6; constructing a written explanation from evidence (NGSS SEP: Analyzing and Interpreting Data, Constructing Explanations, Using Mathematics and Computational Thinking).
Attitudes	Appreciate the use of similarity and enlargement in real-life situations such as vendors' pots and tins. Values: Responsibility (takes care of the tins and cubes borrowed from the environment); Respect (listens to the reasoning of peers). Core competencies: Critical thinking and problem solving (interprets data and infers a rule); Communication and collaboration. PCI: Financial literacy (comparing price per serving) and Environment and technology (reusing tins and boxes).
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	Lesson 5 showed that area scales by k^2 (paint). Lesson 6 completes the puzzle on capacity: volume scales by k^3 , so the large sufuria ($k = 2$) holds $2^3 = 8$ times the small one, about 48 servings against 6. This sets up Lesson 7 (choosing between k , k^2 and k^3 in problems) and Lesson 8 (the final explanation and project).
Safety Notes	Handle tins and boxes with no sharp edges; flatten or file any cut cardboard or tin rims. Do not use real hot githeri; use dry maize or beans, or sand/rice, for any filling. Keep floors clear of spilled grains to prevent slips. Return borrowed tins to their owners.

B. LESSON OVERVIEW

Mama Achieng's large sufuria is 2 times as tall and 2 times as wide as the small one and costs only twice as much (KSh 120 against KSh 60), yet families in Kisumu say it feeds far more than twice as many people. In Lesson 5 learners found that paint area scales by k^2 (KSh 20 to KSh 80). Now they return to their Lesson 1 serving predictions and test them. Groups of 4 first build and measure cubes (side 5 cm and 10 cm) and a flour-tin style cuboid (5 x 4 x 3 cm to 10 x 8 x 6 cm), then compare volumes and discover that doubling every length gives 8 times the volume. They link the rule k^3 to the three lengths in $V = \pi r^2 h$ and apply it to the sufurias (2411.52, 8138.88 and 19292.16 cm³; about 6, 20 and 48 servings of 400 cm³). They then update the DQB and the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Area and Volume	Learners recall Mama Achieng's three sufurias at Kibuye Market, Kisumu. Individually, on a sticky	Hold up the small and large sufuria models (or a picture) and say: "Mama Achieng's large sufuria is	Prediction with reasoning (commit before evidence). Writing a number and a reason exposes the	Collect the sticky notes. Indicators: (1) learners who predict exactly 2 times show the linear

of Similar Solids Principles Source: CK-12 Search ARES for similar videos  HTML: Area and Volume of Similar Solids (Flexbook) Source: CK-12 Search ARES for similar readings	note, they write their Lesson 1 prediction of how many times more githeri the large sufuria (twice as tall and twice as wide) holds than the small one (S about 6 servings). Then they predict, with a quick reason, the volume of a 10 cm cube compared with a 5 cm cube: 2 times, 4 times, 8 times or another number. In pairs they decide which they think is right and sketch the two cubes with a note on why. Predictions are kept visible on the wall for later comparison.	twice as tall and twice as wide. Last lesson we found the paint area goes up 4 times. What do you think happens to the githeri it holds?" Give silent WAIT TIME of 15 seconds before anyone writes. Ask learners to write their prediction before any talk. Then say: "Turn to your partner and compare in 1 minute." Cold-call 4 pairs with: "What number did you choose and why?" Record all answers (2, 4, 6, 8, others) on the board without saying which is right. Say: "Wrong guesses are useful today because we will test them with evidence." Do not reveal 8.	'doubling lengths doubles everything' idea, creating cognitive conflict later and giving learners ownership of what they test.	misconception; (2) learners who predict 4 times may be over-generalising k^2 from Lesson 5; (3) learners who mention three directions (length, width, height) are already reasoning toward k^3. Tally on the board: how many 2, 4, 8, other? Use the tally to decide which groups need extra questioning in the Observe phase.
Observe Phase  VIDEO: Area and Volume of Similar Solids Principles Source: CK-12 Search ARES for similar videos  EXERCISE: Area and Volume of Similar Solids Practice Source: CK-12 Search ARES for similar readings	In groups of 4, learners use ready-made or self-made cubes (from cardboard, small boxes or wooden blocks) and flour-tin style cuboids. Task 1: measure and calculate the volume of a cube of side 5 cm (125 cm³) and a cube of side 10 cm (1000 cm³); work out the ratio $1000 / 125 = 8$ and the linear scale factor $k = 10 / 5 = 2$. Task 2: find the volume of a cuboid $5 \times 4 \times 3$ cm (60 cm³) and its enlargement with $k = 2$, $10 \times 8 \times 6$ cm (480 cm³); ratio $480 / 60 = 8$. Optionally they build the 10 cm cube from eight 5 cm cubes (2 layers of 2×2) or fill the small box with dry maize or sand and pour it into the large one to count how many fills are needed. Groups record all lengths, volumes, k and the volume ratio in a table with the columns: object, k, length ratio, area ratio (from Lesson 5), volume ratio. They add a third row with a quick calculation for $k = 3$ on a cube of side 5 cm (side 15 cm, volume 3375 cm³, ratio 27) to test their emerging rule.	Distribute materials and say: "Check the units: lengths in cm, volume in cm³. Show me every multiplication." Circulate with a checklist. Ask: "What is the ratio of the lengths? What is the ratio of the volumes? What do you notice?" Pose to a struggling group: "How many small cubes of side 5 cm fit along the length, the width and the height of the big cube?" Give WAIT TIME of 10 seconds after each question. Cold-call 3 groups to share their ratio for the cuboid. Ask: "If $k = 2$ gave 8 and Lesson 5 gave 4 for area, what number multiplied by itself gives 8? What do you get with $2 \times 2 \times 2$?" Do not give the rule; ask groups to state it themselves. For the $k = 3$ row say: "Predict first, then calculate. Does your pattern work?" Remind learners to handle tins carefully and to return them.	Data table pattern-finding. Putting k, k^2 and the volume ratio in one table lets learners see the pattern 2, 4, 8 and 3, 9, 27, generalise it, and then test the generalisation with a new case ($k = 3$), which builds a justified rule instead of a memorised formula.	Check each group's table. Indicators: correct volumes 125, 1000, 60, 480 cm³; ratio 8 for both pairs; correct $k = 2$; a stated pattern linking k to the volume ratio; a correct $k = 3$ prediction of 27. Listen for the words 'three lengths' or 'length times width times height'. Mark a group as secure if a member can explain $2 \times 2 \times 2 = 8$ using the three dimensions. Reteach with the layers of small cubes for any group that adds ($2 + 2 + 2 = 6$) instead of multiplying.

Explain Phase  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Area and Volume of Similar Solids (Flexbook) Source: CK-12  Search ARES for similar readings	Learners use their data to explain the sufuria puzzle. Step 1: state the rule in their own words and then in symbols: volume scale factor = k^3 (image volume / object volume). Step 2: link the rule to $V = \pi r^2 h$, noting that capacity depends on three lengths (r, r and h), each multiplied by k. Step 3: calculate with $\pi = 3.14$: $S = 3.14 \times 64 \times 12 = 2411.52 \text{ cm}^3$ (about 2.4 litres); $L = 3.14 \times 256 \times 24 = 19292.16 \text{ cm}^3$ (about 19.3 litres); check $19292.16 / 2411.52 = 8$ and say that L holds exactly 8 times S. M ($k = 1.5$): 8138.88 cm^3, ratio $1.5^3 = 3.375$. Step 4: servings of 400 cm^3: S about 6, M about 20, L about 48 (which is 8 times 6). Step 5: compare with the price: L costs KSh 120 (2 times S) but gives 8 times the githeri, so per serving it is KSh 2.50 against KSh 10 for S. Step 6: each group writes a 3-sentence explanation for a Kisumu customer: 'The large sufuria is $k = 2$ times as tall and wide, so its capacity is ... times, the paint area is ... times, the price is ... times.' They compare with their original sticky-note predictions and correct them.	Ask: "Why does the volume use three lengths while the paint area used only two?" and give 15 seconds of silent WAIT TIME. Cold-call 3 learners to answer and then ask a fourth: "Do you agree or disagree, and why?" Say: "Show your 8 using the sufuria numbers, not just the cubes." Have groups check $19292.16 / 2411.52$ on a calculator. Ask: "Mama Achieng's customers said 'far more than twice'. Whose prediction on the wall was closest? What was missing from the others?" Press for reasons: "Which of the three lengths did the 'twice' prediction forget?" Ask: "Is the large sufuria a bargain for the customer? Is the paint a bargain for Mama Achieng?" (use KSh 80 against KSh 20 from Lesson 5 only as a reminder). Make sure the phrase 'exactly 8 times S' is used rather than 'exactly 19.2 L'. Do not re-teach area; just refer to $k^2 = 4$ from Lesson 5.	Claim-Evidence-Reasoning (CER). Learners make the claim (volume $\times k^3$), cite evidence (cube, cuboid and sufuria data) and give reasoning (three dimensions each multiplied by k). Structuring the argument makes them justify the rule and apply it to the phenomenon.	Read each group's 3-sentence explanation. Indicators: states $k = 2$; states capacity $\times 8$ (exactly 8 times S), about 48 servings against 6; gives reasoning with three lengths; compares correctly with price (KSh 120 for 8 times the food, KSh 2.50 and KSh 10 per serving). Common errors to listen for: saying capacity $\times 6$ or $\times 4$; saying '$k \times 3$'; giving 8 for the area. Use a thumbs-up check: 'Show me 1 to 4 fingers: how confident are you in explaining k^3?' and follow up with fingers 1 and 2.
Driving Question Board (DQB) Creation  VIDEO: Volume in unit cubes by decomposing shape <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES	Learners return to the DQB (class chart with the driving question: Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object?). Each group writes on sticky notes: (a) one question that is now answered (e.g. 'Why does the large sufuria feed about 48 people instead of 12?') and moves it to the 'Answered' column with the evidence; (b) one new question	Say: "Look at the DQB. Which questions can we answer now? What evidence do we have?" Give WAIT TIME of 10 seconds. Cold-call 3 groups to read an answered question and justify it with a number. Ask: "What is the next puzzle?" and cluster the new questions into 'working backwards from area or volume to k', 'real-world prices and maps' and 'making our own similar containers'. Tell learners: "Lesson 7 will give you problems that mix k, k^2 and k^3, and Lesson 8 asks	Question tracking on the DQB. Moving questions from 'open' to 'answered' with evidence shows the growth of understanding and keeps the next questions linked to the phenomenon.	Observe the quality of the questions. Indicators: at least one group moves a question to 'Answered' with k^3 evidence; at least two new questions about reversing the calculation or applying it to other objects; questions that show they see that area and volume behave differently from length. Note groups whose new questions are only about arithmetic and prompt: 'How could this connect to Mama Achieng's business?'

for similar videos EXERCISE: Area and Volume of Similar Solids Practice Source: CK-12 Search ARES for similar readings	(e.g. 'If a photo is enlarged $k = 3$, how much more paper does it need?', 'Can I find k if I only know the volumes?', 'What if the tin is not similar?'). They place new questions on the board and link them to the lessons that may answer them.	you to build and test your own pair of containers." Do not answer the new questions yet.		
Model Building Phase VIDEO: Volume with fractional cubes <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners revise the class model (a written and diagram explanation of why capacity and paint do not just double). In groups they update a poster with: (1) a sketch of S, M and L marked with $k = 1, 1.5$ and 2; (2) a table with columns for length (k), paint area (k^2), capacity (k^3) with the values for L: 2, 4, 8 (and for M: 1.5, 2.25, 3.375); (3) the actual numbers: capacity 2411.52, 8138.88, 19292.16 cm³; servings about 6, 20, 48; (4) a sentence connecting the numbers to the price: price KSh 120 follows $k = 2$ only, so the customer gets about 48 servings (KSh 2.50 each) while Mama Achieng's paint costs about 4 times more (KSh 80 against KSh 20). They add a short gap note: 'We still need to practise working backwards and choosing the right factor.' Each group displays its poster and one learner explains it in 30 seconds during a gallery walk.	Say: "Add what we learned today to the model. Your model must explain both the paint and the githeri." Provide a prompt on the board: 'Length $\times \dots$, area $\times \dots$, volume $\times \dots$'. Circulate and ask: "Where in your model do I see the three lengths?" Give WAIT TIME of 10 seconds and cold-call 3 groups at the gallery walk to explain one part of the model in 30 seconds. Ask the audience: "What evidence supports that claim?" Finish with an exit ticket: 'A tin is enlarged with $k = 3$. By what number is its volume multiplied? Show how you know.' (answer 27). Collect the tickets. Keep the class model on the wall for Lesson 7.	Cumulative model revision with gallery walk. Combining k, k^2 and k^3 in one model, tied to the sufuria data, gives a complete structure for the phenomenon and prepares learners to apply it in problems.	Check posters and the exit ticket. Indicators: poster shows k, k^2, k^3 correctly for L (2, 4, 8) and M (1.5, 2.25, 3.375); gives capacity and servings (about 6, 20, 48); explains price against paint correctly; exit ticket answer 27 with the working $3 \times 3 \times 3$. Learners who answer 9 or 6 need a reteach with the 5 cm to 15 cm cube. Record who can explain the reasoning, not just quote the rule.

D. TEACHER REFLECTION

Which groups still treated 'twice as big' as 'twice the volume', and what physical model (layers of small cubes, pouring dry maize) helped them move on? Did learners keep k^2 (paint) and k^3 (githeri) separate, or did they swap them? Which students explained the three-length reasoning in their own words? How well did the data table support pattern finding, and would a $k = 3$ or $k = 0.5$ row have helped more? Were all learners able to take part in measuring and calculating, or did a few groups dominate? What changes would make the CER explanation stronger before Lesson 7?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We measured cubes (5 cm and 10 cm sides: 125 and 1000 cm ³) and cuboids (5 x 4 x 3 cm and 10 x 8 x 6 cm: 60 and 480 cm ³). Doubling every length ($k = 2$) made the volume 8 times bigger. We also saw the sufuria capacities of about 2.4 litres, 8.1 litres and 19.3 litres and servings of about 6, 20 and 48.
What did I learn?	For similar solids the volume scale factor is k^3 . Area scale factor is k^2 (Lesson 5) and the linear scale factor is k . Capacity depends on three lengths, so it multiplies by $k \times k \times k$. The large sufuria ($k = 2$) holds exactly 8 times the small one.
How does this explain the phenomenon?	Mama Achieng's large sufuria is 2 times as tall and as wide, so its capacity is $2^3 = 8$ times the small one (about 48 servings against 6), not 2 times. The price (KSh 120) follows only $k = 2$, so customers get a bargain (KSh 2.50 against KSh 10 per serving), while the paint area grows 4 times (KSh 80 against KSh 20).

LESSON 7 (80 minutes (2 x 40-minute lessons)): Which Factor Do I Use? Moving Between k , k^2 and k^3 with Posters, Maps and Mama Achieng's Sufurias

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners choose the correct factor (k , k^2 or k^3) for a length, area or volume problem, work backwards from a given area or volume factor to find k , and use these skills to explain how Mama Achieng's prices and paint costs compare.
Knowledge	Learners will be able to: (a) relate the linear scale factor to the area and volume scale factors of similar objects (SLO f); (b) apply similarity and enlargement to real-life situations such as posters, maps and containers (SLO g); (c) appreciate the use of similarity and enlargement in real-life situations (SLO h). Learners know that lengths multiply by k , areas by k^2 and volumes by k^3 (from Lessons 5 and 6), and that if the area factor or volume factor is known, k is found by taking the square root or cube root.
Skills	Identify whether a quantity is a length, an area or a volume before choosing a factor; calculate with k , k^2 and k^3 ; work backwards using square roots and cube roots (for example, area factor 9 gives $k = 3$, volume factor 8 gives $k = 2$); interpret a map scale of 1 : 50 000; compare cost per serving; communicate reasoning. Core competencies: Critical thinking and problem solving; Communication and collaboration. NGSS-aligned practices embedded: Using Mathematics and Computational Thinking; Constructing Explanations; Engaging in Argument from Evidence.
Attitudes	Values: Responsibility (learners take care of the sufuria models and measuring tools) and Respect (learners listen to others' reasoning when groups disagree). Learners appreciate that a single scale factor can explain posters, maps and cooking pots. PCI: Financial literacy (comparing price, paint cost and cost per serving to make informed choices).
Key Inquiry Question	How is similarity and enlargement applied in real life? Sub-question: When I know k , which factor do I use, and when I know an area or volume factor, how do I find k ?
Purpose in Storyline	Lesson 7 of 8. In Lessons 5 and 6 learners found that area scales by k^2 and volume by k^3 and used them to explain the sufuria paint area and capacity. Today they move flexibly between k , k^2 and k^3 in new contexts (poster, map, sufuria costs), and work backwards from a given factor to k . This lets the class revise the model so it can predict cost and capacity for any size, and prepares for the Lesson 8 project and final explanation.
Safety Notes	Cardboard or real sufuria models: handle with care, and keep edges of aluminium sufurias away from hands and faces. Keep walkways clear while groups move. If using phones or tablets for calculators, keep them on desks and follow online safety rules. No food is eaten during the lesson; use only paper or cardboard as stand-ins for githeri.

B. LESSON OVERVIEW

Learners return to Mama Achieng's sufurias at Kibuye Market, Kisumu. The large sufuria (L) is 2 times as tall and wide as the small (S) and costs KSh 120 against KSh 60, yet it holds 8 times as much (about 48 servings against 6) and needs 4 times the paint (about KSh 80 against KSh 20). Learners first predict which factor to use in three quick scenarios, then gather evidence by working through three problem stations: (1) a 10 cm by 15 cm photo enlarged $k = 3$ to a 30 cm by 45 cm poster (area 150 to 1350 cm^2 , ratio 9), (2) a map with scale 1 : 50 000 where 1 cm is 500 m and 1 cm^2 is 250 000 m^2 (25 hectares), and (3) the sufuria price, paint and per-serving comparison (KSh 120, KSh 80 paint, KSh 2.50 vs KSh 10 per serving). They then explain a decision rule for choosing k , k^2 or k^3 and for working backwards from an area or volume factor to k . The class adds to the DQB and revises the model so that it can predict cost and capacity for any similar size. No new ideas are taught beyond Lessons 5 and 6; today is about applying and connecting them.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(About 10 minutes) Learners sit in groups of 4 with the three sufuria models (S, M, L) on display. Each learner silently reads three quick scenarios on the board and writes a prediction in their book before talking: (1) 'A 10 cm by 15 cm photo is enlarged so every length is 3 times longer. By what number does the area multiply?' (2) 'On a map with scale 1 : 50 000, one square centimetre of the map shows how much ground?' (3) 'Mama Achieng's L sufuria costs KSh 120 and S costs KSh 60. Will her paint cost for L be about KSh 120, KSh 80 or KSh 40 if S paint costs about KSh 20?' For each, learners write which factor they think is needed (k , k^2 or k^3) and why, with a number guess. Groups then share and record their group prediction on a sticky note: 'We think ____ because ____'. Learners keep the notes for comparison in the Explain Phase.	Say: 'Look at the three sufurias. Everything about L looks doubled. Today you will be detectives deciding which multiplier to use. First, predict on your own.' Give 2 minutes of silent individual writing. WAIT TIME: wait 10-15 seconds after each question before taking any answer. Then cold-call 4 learners (one per scenario, plus one extra for the reasoning behind a different guess) with: 'What did you write, and what made you choose that factor?' Do not say if predictions are right or wrong. Say: 'Interesting, we have different guesses. Keep your notes safe. We will test them.' Circulate and note common ideas such as 'area is 3 times bigger because $k = 3$ ' and 'L paint costs double because it is double the size'. Record these on the board under the heading 'Ideas to test'.	Predict-Observe-Explain with a Prediction Gallery: learners commit to a prediction and a reason before seeing evidence, so that later evidence has something to confirm or challenge. Naming a factor (k , k^2 or k^3) rather than just a number forces them to link the quantity type to the dimension count, which builds the reasoning needed for the problems.	Listen for learners who (a) name k for length but give no rule for area or volume, (b) choose k for area, or (c) already say 'k squared for area'. Observable indicators: Does the written prediction include a factor name and a reason? Tally on a clipboard how many groups chose k^2 for the photo and how many chose k for the paint cost. Learners who mix up factors are grouped with partners who gave different reasoning during the Observe Phase.
Observe Phase  VIDEO: Scale drawing: centimeters to kilometers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reading Maps, Legends/Scale Practice Source: CK-12  Search ARES	(About 25 minutes) Groups rotate through three problem stations, about 7 minutes each, recording evidence in a table headed 'Quantity Is it a length, area or volume? Factor used Answer'. STATION 1 - Photo and poster: Learners measure a 10 cm by 15 cm photo (area 150 cm ²) and a 30 cm by 45 cm poster (area 1350 cm ²), find $k = 3$ from the lengths, then divide 1350 by 150 and compare with 3 ² . They then work backwards: 'A different poster has 4 times the area of its photo. What is k ?' ($k = 2$, since $2^2 = 4$). STATION 2 - Map: Using the scale	Before rotation say: 'At each station, first ask yourself: is this a length, an area or a volume? Write it down BEFORE you calculate.' Place a data card and rulers at each station. Circulate and ask each group at least once: 'How do you know that is the area factor and not the length factor?' and 'If I told you the volume factor is 27, how would you get k ?' WAIT TIME: wait 10-15 seconds after every question before giving a hint. At Station 2 say: 'Careful with units. 1 cm is 500 m, but is 1 cm ² equal to 500 m ² ? Check with a square drawn on squared paper.' If a	Data Table with a 'Type of quantity' column (structured evidence collection): by forcing learners to classify each quantity before calculating, the strategy builds the habit of matching the factor to the number of lengths involved. Working backwards (square root and cube root) is presented as the same relationship read in reverse, which strengthens understanding of k^2 , k^3 rather than memorising procedures.	Check each group's table. Indicators of understanding: (a) correctly labels at least 5 of 6 quantities as length, area or volume; (b) obtains 9 for the photo area ratio and links it to 3 ² ; (c) gets 25 hectares per map cm ² and 100 hectares for the 4 cm ² farm; (d) gets KSh 2.50 vs KSh 10 per serving; (e) works backwards to $k = 2$ (area factor 4) and $k = 3$ (volume factor 27). Common errors to note: multiplying area by 3 instead of 9, writing 500 m ² for 1 cm ² of map, and cube-rooting an area factor. Record names on a checklist for the Explain Phase.

for similar readings	1 : 50 000 they show that 1 cm on the map is 50 000 cm = 500 m on the ground, so $k = 50\,000$ from map to ground. They compute that 1 cm² on the map is 250 000 m² (25 hectares). They then answer: 'A farm on the map measures 4 cm². How many hectares is it?' ($4 \times 25 = 100$ hectares) and 'A road is 3 cm on the map. How long is the real road?' ($3 \times 500 = 1500$ m). STATION 3 - Sufuria costs: Using the data card (S: KSh 60, about 6 servings, paint about KSh 20; L: KSh 120, about 48 servings, paint about KSh 80), learners compute price per serving (S: $60 / 6 = \text{KSh } 10$; L: $120 / 48 = \text{KSh } 2.50$), the servings factor $48 / 6 = 8 = 2^3$, and the paint cost factor $80 / 20 = 4 = 2^2$. They then work backwards: 'Mama Achieng makes an extra-large sufuria that holds 27 times as much as S. What is k? What would its paint cost be?' ($k = 3$, paint factor 9, so about $9 \times \text{KSh } 20 = \text{KSh } 180$). Learners also check that the real sufuria models agree: height 12 to 24 cm, diameter 16 to 32 cm, $k = 2$.	group is stuck on working backwards, ask: 'What number multiplied by itself gives 9? What number multiplied by itself three times gives 8?' Do not give the answer. At the end of each rotation call 'Switch!' and signal a 1-minute warning. Keep watch on handling of the sufuria models.		
Explain Phase  VIDEO: Making a scale drawing Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(About 20 minutes) Groups return to their Predict sticky notes and compare them with their evidence tables. Each group completes the sentence frames: 'We predicted _____. The evidence showed _____. So the factor for a length is _____, for an area is _____ and for a volume is _____.' Groups then build a 'Decision Rule Card' for choosing a factor, for example: 'Ask: how many lengths are multiplied in this quantity? One length: use k. Two lengths: use k^2. Three lengths: use k^3. If I know the area or volume factor, k is the square root or cube root.' They then use the card to	Say: 'Pull out your first predictions. Which ones changed? Why?' Give 3 minutes of silent individual reflection, then 5 minutes for group discussion. Use the talk move: 'Who can restate what Amina just said in your own words?' Cold-call 3 groups to share the Decision Rule Card, and ask a different learner each time to explain WHY (for example, 'Why does a paint area need two lengths?'). WAIT TIME: 10-15 seconds before each cold call. Address misconceptions with questions, not corrections: 'Mama Achieng's price doubled but paint cost multiplied by 4. Is her	Claim-Evidence-Reasoning with a Decision Rule Card: groups state a claim (which factor to use), support it with numbers from the stations (9 for photo area, 25 hectares per map cm², 8 for servings, 4 for paint) and give reasoning (number of lengths in the quantity). Peer critique with 'agree, disagree or add on' builds scientific argumentation and exposes errors in reasoning rather than just in calculation.	Observable indicators: (a) the Decision Rule Card links the factor to the number of lengths; (b) the explanation for Mama Achieng includes both the per-serving comparison (KSh 2.50 vs KSh 10) and the paint comparison (about KSh 80 vs KSh 20); (c) the exit-ticket answers are 9 and 27. Sort slips into three piles: correct with reasoning, correct without reasoning, incorrect. Plan a short follow-up the next day with the third pile using a smaller k such as 2 and a cube of side 5 cm.

	explain Mama Achieng's situation in 3-4 sentences: why L at KSh 120 gives much cheaper githeri per serving (KSh 2.50 vs KSh 10) while her paint cost rises faster than her price (about KSh 80 against KSh 20, 4 times, but price only 2 times). Two groups present; the rest evaluate with 'agree, disagree or add on'. Finally, learners solve one individual check problem: 'A small tin of cooking oil is similar to a large tin. The large tin is 3 times as tall. (a) What is the ratio of their label areas? (b) What is the ratio of their volumes?' (9 and 27).	large sufuria earning her more or less per shilling of paint?' Press for evidence: 'What number in your table supports that?' Do not re-teach the derivation of k^2 or k^3; if a group is unsure, refer them to their Lesson 5 and 6 notes. Collect the individual check answers on slips as an exit ticket.		
Driving Question Board (DQB) Creation  VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reading Maps, Legends/Scale Practice Source: CK-12  Search ARES for similar readings	(About 10 minutes) Learners revisit the class Driving Question Board, which displays the driving question: 'Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object?' Each group reviews earlier question cards and moves those they can now answer to an 'Answered' column, attaching a sticky note with the evidence, for example 'If the volume factor is 27, then $k = 3$'. Each learner writes at least one new question on a card, such as 'Does this work for a sufuria with a lid?', 'What if k is less than 1, like a tiny toy sufuria?', or 'Can Mama Achieng choose a size where paint cost equals price?' Groups post cards under headings: 'Answered', 'Partly answered' and 'New questions for Lesson 8'.	Say: 'Walk to the board with your group. Which questions can you now answer with evidence from today?' Give 2 minutes for the group to move cards before whole-class review. WAIT TIME: 10-15 seconds before calling on anyone. Cold-call 3 learners to read one answered question and justify it with a number from their table. Say: 'Which new question could we test in our project next lesson?' Choose 2 new questions to link to Lesson 8, for example the question about reductions ($0 < k < 1$) and the design of similar containers. Gently reject questions that need negative scale factors, as they are outside this sub-strand, with: 'Great curiosity, but we will keep to positive factors this term.' Photograph the board for the record.	Driving Question Board as a living record of understanding: moving questions from 'unanswered' to 'answered' with evidence shows learners how their knowledge grows. Writing new questions from new gaps builds ownership and gives the next lesson a purpose.	Indicators: at least one question card per learner; at least half of the groups attach correct evidence to an answered card; new questions show deeper thinking (for example, about reductions, cost per serving or other shapes) rather than repeating earlier ones. Note learners who write no question or only recall questions, and follow up individually.
Model Building Phase	(About 15 minutes) Groups revise the class model of 'why capacity and paint do not just double'. In	Say: 'Your model must now predict, not just describe. Use it to answer the $k = 3$ question without	Cumulative Model Revision with prediction testing: the model becomes a tool for prediction when	Indicators: (a) the model includes k, k^2 and k^3 with matching examples from the sufurias; (b) the

 VIDEO: Scale drawing: centimeters to kilometers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	their model sheets (diagram plus words), they add a new section titled 'Predicting any size'. For the sufuria S to L ($k = 2$), learners show a table: Length $k = 2$; Paint area $k^2 = 4$ (about KSh 20 to KSh 80); Capacity $k^3 = 8$ (about 6 to 48 servings); Price KSh 60 to KSh 120 (follows $k = 2$). They add the new rule from today: 'To go backwards, area factor 9 means $k = 3$; volume factor 27 means $k = 3$'. They test the model by predicting for a new size without measuring: 'A sufuria with $k = 3$ from S. How many servings and how much paint?' (volume factor 27, so about $6 \times 27 = 162$ servings; paint factor 9, so about KSh 180). They also state one limit of the model: price per serving falls for larger sizes (KSh 10 to KSh 2.50) while paint costs per serving fall less steeply, so the vendor's choices depend on both. Groups compare their model to the Predict notes and mark what changed.	touching a ruler.' Provide model sheets and markers. WAIT TIME: 10-15 seconds after posing each prediction question before taking answers. Circulate and ask: 'Where in your model is the reason that paint multiplies by 4 but capacity by 8?' and 'What does your model say about price compared with capacity?' Cold-call 2 groups to share their $k = 3$ prediction and the reasoning. Remind learners: 'Use 'about' for rounded values and keep to $\pi = 3.14$ when you recalculate.' Collect the models. Preview Lesson 8: 'Next time you will build your own pair of similar containers and test whether your model predicts their area and volume.'	learners use it for a new size ($k = 3$) without measuring. Adding the working-backwards rule and noting a limit of the model builds deeper, more flexible understanding and links the lesson to Mama Achieng's real decisions.	working-backwards rule is stated correctly; (c) the $k = 3$ prediction gives about 162 servings and about KSh 180 paint (or an equivalent factor of 27 and 9); (d) the model links price following k and paint following k^2. Use a 4-point check (all four present = secure; three = developing; two or fewer = needs support) and note groups for targeted support in Lesson 8.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners classify each quantity as a length, area or volume BEFORE calculating? Which groups skipped this step and what happened? (2) Which scenario caused the most errors: the map (units of m and m^2), the poster (k vs k^2) or the sufuria costs (price vs paint vs per-serving)? (3) Could learners work backwards from 9 to $k = 3$ and 27 to $k = 3$ without prompting? Which learners needed the 'what number times itself' hint? (4) Did I keep WAIT TIME of 10-15 seconds, and did quieter learners contribute in the cold calls? (5) Did any learner confuse the price following k with the paint cost following k^2 ? (6) What evidence do I have that the class is ready for the Lesson 8 project, and which groups need support with measuring and calculating for cylinders with $k = 1.5$? Record adjustments, such as extra practice on map units or a smaller second factor like $k = 1.5$ for slower groups.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that when the photo is enlarged with $k = 3$ the area multiplies by 9 (150 cm^2 to 1350 cm^2); on the 1 : 50 000 map 1 cm is 500 m but 1 cm^2 is 250 000 m^2 (25 hectares); and for Mama Achieng's sufurias the large one ($k = 2$) costs only 2 times as much (KSh 120 vs KSh 60) but holds 8 times as much (about 48 servings against 6) and needs about 4 times the paint (about KSh 80 against KSh 20), so a serving costs KSh 2.50 instead of KSh 10.
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What did I learn?	We learned that a length uses k , an area uses k^2 and a volume uses k^3 , and that we can work backwards: an area factor of 9 means $k = 3$, and a volume factor of 27 means $k = 3$. We learned to ask first whether a quantity is a length, an area or a volume.
How does this explain the phenomenon?	We can explain that the large sufuria holds far more than twice the githeri because capacity depends on three lengths (radius, radius and height), so it multiplies by $k^3 = 8$, while paint depends on two lengths so it multiplies by $k^2 = 4$, and the price only follows $k = 2$. This is why large customers get a bargain while Mama Achieng's paint costs rise faster than her price.

LESSON 8 (80 minutes (2 x 40-minute lessons, double period)): Project: Make Similar Containers and Give the Final Explanation of the Sufuria Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply all their ideas to design, build and document a pair of similar containers (cube, cuboid or cylinder), predict and check their area and volume, and give the complete final explanation of why Mama Achieng's large sufuria holds 8 times as much githeri and needs 4 times as much paint, while her price only doubles.
Knowledge	By the end of the lesson, the learner should be able to: a) identify properties of similar figures, b) determine the centre of enlargement and the linear scale factor of similar figures, d) determine the area scale factor of similar figures, e) determine the volume scale factor of similar objects, f) relate linear scale factor to area and volume scale factor of similar objects, g) apply similarity and enlargement to real-life situations, h) appreciate the use of similarity and enlargement in real-life situations. Project (KICD): use the knowledge and skills of similarity and enlargement to make similar containers in the shape of cubes, cuboids and cylinders that are of different sizes, using locally available materials.
Skills	Planning and measuring accurately; predicting with k , k^2 and k^3 and checking by calculation (Planning and Carrying Out Investigations; Using Mathematics and Computational Thinking); constructing a written and diagram model; engaging in argument from evidence; documenting a process with photos or video clips on a digital device.
Attitudes	Core competencies: Digital literacy (learners connect and interact with digital technology when using phones or tablets to record the project and learn more on similarity and enlargement); Creativity and Imagination (learners draw and build images of given objects under a chosen scale factor). Value: Responsibility (learners take care of the objects and materials collected from the immediate environment). PCI: Environment and technology (learners sort and manage waste materials such as tins, bottles and packets responsibly after collecting them).
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	This is the final lesson. In Lesson 1 learners predicted that the large sufuria would feed about twice as many people. Lessons 2 to 7 taught similar figures, enlargement, k , k^2 and k^3 and how to choose the right factor. Today learners build their own similar containers to test these ideas first-hand, write the Final Explanation of the sufuria phenomenon, compare it with their Lesson 1 model, and complete the DQB.
Safety Notes	Cut cardboard with blunt-nosed scissors or a craft knife used only by the teacher or under direct supervision; cut away from the body on a flat desk. Check empty tins for sharp rims and file or tape the edges. Use only clean, dry, dead-safe materials (no broken glass, no chemically contaminated containers). Filling tests use dry maize, beans or sand over a tray to avoid spills and slipping. Keep phones and tablets away from the filling area. Wash hands after handling the materials and recycle all offcuts.

B. LESSON OVERVIEW

Mama Achieng at Kibuye Market in Kisumu sells githeri in a small (S), medium (M) and large (L) sufuria. L is 2 times as tall and 2 times as wide as S and costs KSh 120 against KSh 60 for S. Yet her customers say L feeds far more than twice as many people, and her oil paint for the outside costs more than double. In this lesson groups first build a pair of similar containers from local materials: cylinder tins (small tin radius 4 cm and height 10 cm, $V = 502.4 \text{ cm}^3$; larger tin radius 6 cm and height 15 cm, $V = 1695.6 \text{ cm}^3$, $k = 1.5$) or cubes (side 5 cm to side 10 cm, 125 cm^3 to 1000 cm^3). They predict area and volume with k^2 and k^3 , test by calculation and by filling, and document the project with photos or video. Then the class returns to the sufuria phenomenon: capacity depends on three lengths so it multiplies by $k^3 = 8$ (about 48

servings against 6), the paint area depends on two lengths so it multiplies by $k^2 = 4$ (about KSh 80 against KSh 20), and the price only follows k (KSh 120 = 2×60). Learners write the Final Explanation, compare their final model with their Lesson 1 model, and complete the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups of 4 look at the three sufurias S, M, L (or cardboard models) and the two project pairs on the teacher's table: a small and large cylinder tin (radius 4 cm, height 10 cm and radius 6 cm, height 15 cm) and two cubes (side 5 cm and side 10 cm). Each learner writes on a sticky note: (1) the scale factor of each pair, (2) how many times more the bigger one holds, (3) how many times more paint it needs. They then compare these predictions with what they wrote in Lesson 1 about how many people L would feed (most said about twice as many). Groups record their reasoning, e.g. 'the cube is twice as long, so volume is $2 \times 2 \times 2 = 8$ times'.	Display the three sufurias and the project pairs. Say: 'Mama Achieng's large sufuria is twice as tall and twice as wide as the small one. Before we build anything, what do you predict for our own containers?' Give silent think time: WAIT TIME 15 seconds before anyone writes. Then cold-call 4 learners from different groups: 'Which factor will you use for the volume, and why?' Do not correct anything yet; record the range of predictions on the board. Ask 2 learners to read their Lesson 1 prediction about servings aloud and say: 'Keep that number in mind; we will return to it at the end.' Circulate and note who uses k only and who already uses k^2 or k^3 .	Prediction with a stated reason (Predict-Observe-Explain). Writing a number and a reason before testing makes learners commit to their ideas, so the later comparison with data shows how their thinking has changed. Returning to the Lesson 1 prediction links today's project to the original phenomenon.	Observe whether each sticky note has all three parts (k , volume factor, area factor). Indicators: learners who write 'twice as big so holds twice as much' still hold the Lesson 1 misconception; learners who write 8 or 4 with a reason using k are ready. Tally the three types of prediction on the board and use the tally to plan grouping for the build.
Observe Phase  VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum)  Search ARES for similar	Project build and check (about 35 minutes). Each group chooses a pair of similar containers to make from local materials such as cardboard, empty tins, plastic bottles and maize-flour packets. Options: (a) two cubes, side 5 cm and side 10 cm; (b) two cuboids, $5 \times 4 \times 3$ cm and $10 \times 8 \times 6$ cm; (c) two cylinders, radius 4 cm and height 10 cm and radius 6 cm and height 15 cm. Steps: 1) Plan and record the dimensions and k in a table. 2) Measure and cut or select materials accurately. 3) Build both containers and check similarity: all lengths in the same ratio. 4)	Say: 'You are Mama Achieng's container designers. Your job is to build two containers of the same shape and show me with numbers that they are similar.' Give one clear role per learner: measurer, builder, recorder, photographer. Circulate and ask: 'Show me how you know the height and the width are in the same ratio.' WAIT TIME 10 seconds before prompting. For groups that struggle, ask: 'What is image length divided by object length for the radius? Is it the same for the height?' Cold-call 3 groups to state their k and their measured volume ratio. Remind	Evidence gathering by building and measuring (Planning and Carrying Out Investigations). Constructing the objects themselves lets learners see that a ratio of lengths of k gives a ratio of capacities of k^3 . The fill-and-pour test (8 small cubes filling the large cube) gives a concrete picture of the cube factor beyond the formula.	Use an observation schedule. Look for: (1) all corresponding lengths in the same ratio within about 1 mm; (2) a recorded k in the table; (3) volumes calculated correctly (502.4 and 1695.6 cm ³ ; 125 and 1000 cm ³); (4) the volume ratio calculated and compared with k cubed; (5) photographs or clips saved with at least three steps captured. Note groups that report a ratio of 2 or 1.5 for volume; ask them to recalculate.

readings	Calculate the volume and surface area of each. Check cases: cubes 125 and 1000 cm³ (ratio 8); cuboids 60 and 480 cm³ (ratio 8); cylinders $V = 3.14 \times 4 \times 4 \times 10 = 502.4$ cm³ and $V = 3.14 \times 6 \times 6 \times 15 = 1695.6$ cm³ (ratio $1695.6 / 502.4 = 3.375 = 1.5 \times 1.5 \times 1.5$). 5) Fill the small container with dry maize or sand, then pour it into the larger one to see how many fills are needed (cubes: 8 small fills for the large cube). 6) Take photos or short video clips of each step and write down any challenges faced, such as inaccurate measuring, and how they were solved. Save the files on the group's digital device.	learners to photograph the measuring steps, not only the finished product. Check safety: cutting tools, tin rims, and that dry material stays over the tray.		
Explain Phase  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Final Explanation (about 20 minutes). Step 1: Each group fills in a comparison table for its own containers: k, area factor (k^2), volume factor (k^3), measured or calculated ratio. Step 2: Return to the sufuria phenomenon. Using the facts: S has height 12 cm and radius 8 cm; L has height 24 cm and radius 16 cm, so $k = 2$. Capacity $S = 3.14 \times 64 \times 12 = 2411.52$ cm³ (about 2.4 litres) and $L = 3.14 \times 256 \times 24 = 19292.16$ cm³ (about 19.3 litres), exactly 8 times S, which is about 48 servings against 6. Outside area $S = 803.84$ cm² and $L = 3215.36$ cm², exactly 4 times S, so paint costs about KSh 80 against KSh 20. The price follows k only: $KSh\ 60 \times 2 = KSh\ 120$. So per serving, L costs KSh 2.50 against KSh 10 for S. Step 3: Each learner writes the Final Explanation in their own words using this frame: 'The three sufurias are similar because ... $k = \dots$ Capacity depends on ... lengths so it multiplies by ... Paint area depends on ... so it multiplies	Say: 'Now answer the question we started with: why does a sufuria twice as tall and twice as wide hold far more than twice the githeri?' Ask groups to link each part of their explanation to a piece of evidence from today's build. WAIT TIME 15 seconds, then cold-call 4 learners, one for each question: Why 8 for capacity? Why 4 for paint? Why only 2 for the price? Who gains and who loses, the customer or Mama Achieng? Press for reasoning: 'How many lengths does capacity depend on? Name them.' Accept 'radius, radius, height'. Ask: 'What would the large sufuria hold if k were 3?' (27 times). Use the exact wording 'exactly 8 times S', not 'exactly 19.2 litres'. Select two learners to read their paragraph; the class gives one 'star' (clear use of evidence) and one 'wish' (a missing step).	Claim-Evidence-Reasoning with a sentence frame. Learners state the claim (8 times, 4 times, price 2 times), cite evidence from the sufuria data and their own containers, and give the reasoning (number of lengths multiplied: three lengths give k^3, two lengths give k^2, one length gives k). Peer star-and-wish feedback sharpens the explanations.	Check each written explanation against this checklist: (1) states that S, M, L are similar with constant ratio; (2) identifies $k = 2$; (3) links capacity to three lengths and $k^3 = 8$; (4) links paint area to two lengths and $k^2 = 4$; (5) states that the price follows k; (6) gives KSh figures (KSh 80 against KSh 20; KSh 2.50 against KSh 10); (7) uses own container data as an extra example. Score 0 to 7 as an observable indicator; learners with 5 or more are ready for independent application; learners with fewer receive a short teacher conference.

	by ... The price only ... so Mama Achieng's ... while customers ...'. They give one extra example from their own container pair (e.g. 'my cylinders: $k = 1.5$, volume 3.375 times').			
Driving Question Board (DQB) Creation  VIDEO: Dilating shapes: shrinking by 1/2 <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Similarity & transformations <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Class reviews the DQB built across Lessons 1 to 7 (questions such as: Why does the large sufuria feed so many more people? Why does the paint cost more than double? What is a scale factor?). In groups, learners move each sticky-note question to one of three columns: 'Answered' (with a one-line answer and evidence), 'Partly answered', 'New question'. They add at least one new question inspired by the project, for example 'What happens to weight if the tin is made of thicker metal?' or 'How could the price per serving be fair for Mama Achieng?'. They add a final response card to the main driving question: 'Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object?'	Say: 'Let us look at our board from Lesson 1. Which questions can we answer with evidence now?' Use WAIT TIME 10 seconds before the first group moves a note. Cold-call 3 groups to place one question and justify it by pointing to evidence (a lesson, a measurement or a calculation). Challenge: 'What is still not answered? What would you need to find out?' Record the new questions and accept ones that go beyond today (e.g. changing shape, not only size). Ask 2 learners to read the final response card aloud.	Question tracking with the Driving Question Board. Sorting questions into answered, partly answered and new makes learning visible, shows progress from confusion to explanation, and keeps curiosity alive beyond the sub-strand.	Observe whether learners justify placements with evidence and not only opinion. Indicators: at least 80% of Lesson 1 questions moved to 'Answered' with correct evidence; each group adds one productive new question; the final response card correctly mentions k , k^2 and k^3 .
Model Building Phase  VIDEO: Intro to triangle similarity <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Model comparison (about 10 minutes). Learners take out their Lesson 1 model (the first drawing and sentence about how big the large sufuria is and how much it feeds). In pairs they build the final model on a poster or in their books: a labelled diagram of sufuria S (radius 8 cm, height 12 cm) and L (radius 16 cm, height 24 cm) showing the length arrow $k = 2$, the paint area arrow $k^2 = 4$ (803.84 to 3215.36 cm ² ; KSh 20 to KSh 80), the capacity arrow $k^3 =$	Say: 'Place your first model next to your final model. What changed, and what evidence changed it?' Give WAIT TIME 10 seconds before pairs begin. Cold-call 3 pairs to read one row of their Then and Now table. Ask: 'If Mama Achieng made an even larger sufuria with $k = 3$, what would you tell her about paint, githeri and price?' (paint 9 times, githeri 27 times, price 3 times). Invite learners to suggest how the project containers could be stored	Cumulative model revision and model comparison. Placing the Lesson 1 model next to the final model makes the conceptual change explicit; adding the extra example from learners' own containers shows that the model transfers to new objects, not just the sufurias.	Check that the final model: (1) shows all three arrows k , k^2 , k^3 with the correct numbers (2, 4, 8); (2) uses the correct data (803.84 and 3215.36 cm ² ; 2411.52 and 19292.16 cm ³ ; 6 and 48 servings; KSh 20 and 80; KSh 60 and 120); (3) explains each factor by the number of lengths; (4) names the evidence that changed the learner's thinking in the Then and Now table; (5) correctly predicts the $k = 3$ case. Use a rating scale (Exceeds, Meets, Approaches,

 EXERCISE: Use similar triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	8 (2411.52 to 19292.16 cm³; 6 to 48 servings) and the price arrow (KSh 60 to KSh 120). Beside it, a mini-diagram of their own container pair. They then complete a two-column 'Then and Now' table: Lesson 1 model versus Final model (What I thought; What I know now; The evidence that changed my mind). Finally they add the summary table prompt below to the model and display work for a gallery walk with documented photos or video clips of the project.	and used as a learning material in the classroom or maths club. Close with: 'Whenever something is enlarged, ask: how many lengths does this quantity depend on?'	Below) based on the KICD rubric for determining area and volume scale factors.
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D. TEACHER REFLECTION

Did every group build two containers that were truly similar, with all lengths in the same ratio? Which learners still used k for volume or area, and what evidence changed their thinking (the fill-and-pour test, the calculated 3.375, or the sufuria figures)? Were the Final Explanations complete, with all seven checklist points, and did they link to Mama Achieng's price, paint cost and servings? How well did documentation (photos, videos, challenges recorded) support assessment of the project, and how can the containers be stored and reused as a learning resource? What new questions on the DQB can seed future lessons, for example on surface area of solids and volume and capacity?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We built a small and a larger container with all lengths in the same ratio (for example cylinder tins with radius 4 cm and height 10 cm, and radius 6 cm and height 15 cm, $k = 1.5$; or cubes of side 5 cm and 10 cm). We measured them, filled the small one with maize or sand and poured it into the larger one, and recorded the process with photos or video. We also reviewed Mama Achieng's three sufurias: S, M and L, with L twice as tall and twice as wide as S.
What did I learn?	When an object is enlarged by a linear scale factor k , lengths multiply by k , areas multiply by k^2 and volumes multiply by k^3 . Our cylinders: $k = 1.5$, volume 502.4 cm ³ to 1695.6 cm ³ , ratio 3.375 = 1.5^3 . Our cubes: $k = 2$, volume 125 cm ³ to 1000 cm ³ , ratio 8. A quantity depends on as many lengths as it has dimensions: one length, two lengths or three lengths.
How does this explain the phenomenon?	The large sufuria is similar to the small one with $k = 2$. Its capacity depends on three lengths (radius, radius, height), so it holds exactly 8 times as much: 19292.16 cm ³ against 2411.52 cm ³ , about 48 servings against 6. Its outside area depends on two lengths, so it is 4 times as large (3215.36 cm ² against 803.84 cm ²), and paint costs about KSh 80 against KSh 20. The price depends only on the linear factor (KSh 60 $\times$ 2 = KSh 120), so customers get a bargain (KSh 2.50 per serving against KSh 10) while Mama Achieng's paint cost rises faster than her price.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.